



Guided waves in a Timoshenko beam with a bonded composite patch: Frequency domain spectral element modeling and analysis



Injoon Jang, Ilwook Park, Usik Lee *

Department of Mechanical Engineering, Inha University, 253 Yonghyun-dong, Nam-gu, Incheon 402-751, Republic of Korea

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ABSTRACT

Simulation of ultrasonic guided waves in a beam-type structure with a bonded composite patch repair requires a proven modeling and analysis technique that can provide accurate dynamic solutions at a high frequency range. We developed a spectral element model for a metallic beam with a bonded composite patch by using the variational method. To that end, the governing equations of motion are derived by adopting Timoshenko beam theory for the transverse bending motion, and Mindlin–Herrmann rod theory for the longitudinal motion of the composite patched beam to take into account the effects of lateral contraction in the thickness direction. The high accuracy of the spectral element model is then evaluated by comparison with the solutions obtained by the conventional finite element method (FEM) and by the commercial finite element (FE) analysis package ANSYS. By using the spectral element model, the effects of the composite patch on the characteristics of guided waves are numerically investigated. In addition, the effects of debonding are also numerically investigated with respect to monitoring the integrity of the bonded composite patch repair.

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1. Introduction

It is very efficient and economical to use bonded patch repair techniques to repair damaged structural components, or to restore the load-carrying capacity of weakened structures. The bonded composite patch repair (BCPR), which was initiated by Baker [1], has been widely applied to civil and military aircraft because composite materials have obvious advantages including high specific stiffness and strength, longer fatigue life, corrosion and wear resistance, tailoring ability, and ease of fabrication. Extensive reviews on related works up to 2002 are available in Baker et al. [2].

A BCPR reduces the stresses near the damage by bridging between the damaged parent structure and the composite patch in order to retard or completely stop damage growth beneath the composite patch. However, the performance and durability of a BCPR certainly depend on the structural integrity of the repair, which can be significantly degraded by debonding (bond failure) at the interface between the parent structure and composite patch. Many experimental and numerical studies show that peak peeling and shear stresses develop at the edge of the bonded patch such that debonding may occur at the edge regions of the patch [3–5]. Thus, once repairs are done, it is important to inspect or monitor the integrity of bonded patches by using a proper nondestructive testing (NDT) technique or structural health monitoring (SHM) technique [2,6]. BCPR techniques are usually applied to thin-walled

structures such as aircraft skin. Recently, Chiu et al. [7] and Habid et al. [8] used ultrasonic guided Lamb waves to monitor a composite repair patch system.

There have been many studies [9–17] of the stress intensity factors, crack growth rates, and dynamic responses (natural frequencies and frequency response functions) of structures with BCPRs. However, there are few studies of the characteristics of ultrasonic guided waves propagating in thin-walled structures with BCPRs. In previous works [3–5,7,9–17], the finite element method (FEM) has been used for most numerical studies.

In order to successfully investigate ultrasonic guided waves propagating in a thin-walled structure with a bonded composite patch repair, the dynamic coupling between the parent structure and composite patch must be modeled accurately and analyzed in an efficient way. The FEM is a powerful tool for solving diverse and complex engineering problems. However, simple polynomials that are not related to the vibrating frequency are normally used as shape functions to formulate conventional finite element models. Thus, very fine meshes are often needed in order to improve an FEM solution, especially at high frequencies. This could increase the computation time and cost significantly. In contrast to the FEM, the spectral element method (SEM) is known as an exact element method because a spectral element model represented by an exact dynamic stiffness matrix is formulated by using the frequency-dependent dynamic shape functions that are derived from free wave solutions satisfying governing equations in the frequency (ω) domain [18]. The “frequency-domain” SEM can be considered to be an alternative to the FEM because it can provide

* Corresponding author. Tel.: +82 32 860 7318; fax: +82 32 866 1434.

E-mail address: ulee@inha.ac.kr (U. Lee).

extremely accurate dynamic solutions at high frequency by using only a small number of meshes, while reducing computation time and cost significantly.

From the historical point of view, the frequency domain SEM was initiated by Narayanan and Beskos [19], and was later named “SEM” by Doyle [20]. The conventional FEM is a time (t) domain method and it is normally formulated in the form of an ordinary differential equation: $\mathbf{M}\ddot{\mathbf{d}}(t) + \mathbf{C}\dot{\mathbf{d}}(t) + \mathbf{K}\mathbf{d}(t) = \mathbf{f}(t)$, where $\mathbf{M}$, $\mathbf{C}$, and $\mathbf{K}$ are the finite element mass matrix, damping matrix, and stiffness matrix, respectively; $\mathbf{d}$ is the nodal degrees of freedom (DOFs) vector; and $\mathbf{f}$ is the nodal forces vector. On the other hand, the frequency domain SEM is formulated in the form of an algebraic equation such as $\mathbf{S}(\omega)\mathbf{d}(\omega) = \mathbf{f}(\omega)$, where $\mathbf{S}(\omega)$ is the exact dynamic stiffness matrix called “spectral element matrix”, $\mathbf{d}(\omega)$ is the spectral nodal DOFs vector, and $\mathbf{f}(\omega)$ is the spectral nodal forces vector. Although the “time domain” SEM described by Patera [21] is also often called “SEM” in the literature, it is completely different from the frequency domain SEM because it belongs to the p -version FEM where the higher order polynomials such as orthogonal Chebyshev polynomials are used as the shape functions.

Despite the advantages of SEM, especially when applied to one-dimensional (1-D) problems, there have been few applications to multi-layer structures, including patch-repaired beams and smart beam structures. Lee and Kim [22] and Lee et al. [23] were the first to apply SEM to a smart elementary beam that consists of a metallic base beam and a surface-bonded piezoelectric (PZT) layer, and recently to a smart composite Timoshenko beam consisting of a composite base beam and a surface-bonded PZT layer. To the authors’ best knowledge, SEM has not yet been applied to structures with a surface-bonded composite patch.

The goals of this study are (1) to derive governing equations of motion and associated boundary conditions for beams with a bonded composite patch (hereafter called composite patched beams); (2) to formulate a spectral element model by using the variational approach; (3) to evaluate the accuracy of the spectral element model by comparison with other solution methods; and (4) to investigate the effects of patch-edge debonding on the characteristics of guided waves with respect to BCPR health monitoring applications.

2. Derivation of the equations of motion

2.1. Kinematics

Fig. 1 shows a composite patched beam that consists of two layers: the base beam layer (simply, the base beam) of thickness h_b and the laminated composite patch layer (simply, the composite patch) of thickness h_c . We assumed that the composite patch is perfectly bonded on the whole upper surface of the base beam. The composite patched beam has length L , width b , and total thickness $h = h_b + h_c$. We also assumed that the composite patched beam has motion only in the (x, z)-plane as shown in Fig. 1(b). By adopting Timoshenko beam theory [24] to take into account the shear deformation and rotary inertia effects, and Mindlin–Herrmann rod theory [25] to take into account the effect of the lateral contraction in the z -direction, the axial (longitudinal) and transverse displacements of the base beam can be represented as follows:

$$\begin{aligned} u_b(x, z, t) &= u_{b0}(x, t) - z\theta_b(x, t) \\ w_b(x, z, t) &= w_{b0}(x, t) + z\psi_b(x, t) \end{aligned} \quad (1)$$

where $u_{b0}(x, t)$ and $w_{b0}(x, t)$ are the axial and transverse displacements on the mid-plane of the base beam, respectively; $\theta_b(x, t)$ is the rotation of the cross-section normal to the mid-plane of the base beam about the y -axis; $\psi_b(x, t)$ is the lateral contraction; t is the time; and z is measured with respect to the mid-plane of the

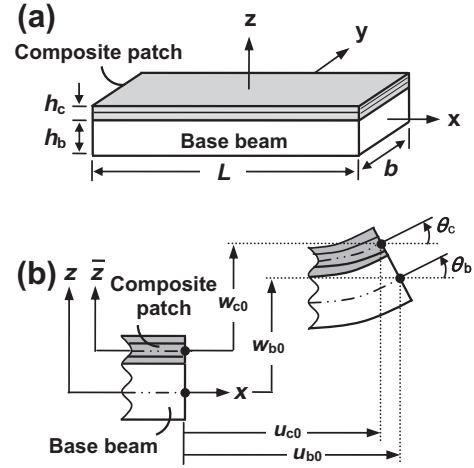


Fig. 1. Composite patched beam: (a) geometry and (b) deformations.

base beam. Similarly, the displacement fields in the composite patch can be represented by

$$\begin{aligned} u_c(x, \bar{z}, t) &= u_{c0}(x, t) - \bar{z}\theta_c(x, t) \\ w_c(x, \bar{z}, t) &= w_{c0}(x, t) + \bar{z}\psi_c(x, t) \end{aligned} \quad (2)$$

where $\bar{z}$ is measured with respect to the mid-plane of the composite patch.

Based on the assumption that the composite patch is perfectly bonded on the upper surface of the base beam, the displacement connectivity conditions can be imposed on the interface between the base beam and composite patch to obtain the following relations:

$$\begin{aligned} u_{c0}(x, t) &= u_{b0}(x, t) - \frac{1}{2}h_b\theta_b(x, t) - \frac{1}{2}h_c\theta_c(x, t) \\ w_{c0}(x, t) &= w_{b0}(x, t) + \frac{1}{2}h_b\psi_b(x, t) + \frac{1}{2}h_c\psi_c(x, t) \end{aligned} \quad (3)$$

2.2. Equations of motion

The equations of motion and the relevant boundary conditions for the composite patched beam can be derived by using Hamilton's principle, given by [24]

$$\int_{t_1}^{t_2} (\delta T - \delta V + \delta W) dt = 0 \quad (4)$$

where T is the kinetic energy, V is the strain energy, and δW is the virtual work done by external loadings.

The strain energy in a composite patched beam of length L can be obtained by summing the strain energy in the base beam (Eq. (A7) in Appendix A) and the strain energy in the composite patch (Eq. (B6) in Appendix B) as follows:

$$\begin{aligned} V &= \frac{1}{2} \int_0^L \{ A_b C_{11}^* u_{b0}^2 + I_{ob} Q_{11}^* \theta_b^2 + 2A_b C_{13}^* u_{b0}' \psi_b \\ &\quad + \kappa_b Q_{55}^* A_b (w_{b0}' - \theta_b)^2 + \kappa_b C_{55}^* I_{ob} \psi_b^2 + A_b C_{33}^* \psi_b^2 \} dx \\ &\quad + \frac{1}{2} \int_0^L \{ bA_{11} u_{c0}^2 + b\bar{D}_{11} \theta_c^2 + 2bA_{13} u_{c0}' \psi_c + \kappa_c b\bar{A}_{55} (w_{c0}' - \theta_c)^2 \\ &\quad + \kappa_c bD_{55} \psi_c^2 + bA_{33} \psi_c^2 \} dx \end{aligned} \quad (5)$$

The symbols used in Eq. (5) are defined in Appendices A and B, and the subscripts b and c denote quantities for the base beam and composite patch, respectively. By using the displacement fields given by Eqs. (1) and (2), the kinetic energy in the composite patched beam is given by

$$T = \frac{1}{2} \int_0^L \left[\rho A_b \{ \dot{u}_{b0}^2 + \dot{w}_{b0}^2 \} + \rho I_{ob} \{ \dot{\theta}_b^2 + \dot{\psi}_b^2 \} \right] dx + \frac{1}{2} \times \int_0^L \left[\rho A_c \{ \dot{u}_{c0}^2 + \dot{w}_{c0}^2 \} - 2\rho R_c (\dot{u}_{c0}\dot{\theta}_c - \dot{w}_{c0}\dot{\psi}_c) + \rho I_{oc} (\dot{\theta}_c^2 + \dot{\psi}_c^2) \right] dx \quad (6)$$

where

$$\begin{aligned} \{\rho A_b, \rho I_{ob}\} &= \int_{-h_b/2}^{h_b/2} b \{\rho_b, \rho_b z^2\} dz \\ \{\rho A_c, \rho R_c, \rho I_{oc}\} &= \int_{-h_c/2}^{h_c/2} b \{\rho_c, \rho_c z, \rho_c z^2\} dz \end{aligned} \quad (7)$$

and where ρ_b and ρ_c denote the mass densities of the base beam and composite patch, respectively. Finally, the virtual work done by the external forces and moments applied to the base beam and composite patch can be obtained as

$$\begin{aligned} \delta W &= N_{b1} \delta u_{b0}(0) + N_{b2} \delta u_{b0}(L) + M_{b1} \delta \theta_b(0) + M_{b2} \delta \theta_b(L) \\ &+ Q_{b1} \delta w_{b0}(0) + Q_{b2} \delta w_{b0}(L) + \int_0^L \{ f_{bu}(x, t) \delta u_{b0} \\ &+ f_{bw}(x, t) \delta w_{b0} + f_{b\theta}(x, t) \delta \theta_b \} dx + N_{c1} \delta u_{c0}(0) \\ &+ N_{c2} \delta u_{c0}(L) + M_{c1} \delta \theta_c(0) + M_{c2} \delta \theta_c(L) + Q_{c1} \delta w_{c0}(0) \\ &+ Q_{c2} \delta w_{c0}(L) \end{aligned} \quad (8)$$

where N_i , M_i , and Q_i ($i = 1, 2$) are the resultant axial forces, bending moments, and transverse shear forces applied at the boundaries $x = 0$ and L , and $f_{bu}(x, t)$, $f_{bw}(x, t)$, and $f_{b\theta}(x, t)$ are the external loadings distributed along the x -axis.

By substituting Eqs. (5), (6) and (8) into Eq. (4), followed by some mathematical manipulation, six fully-coupled governing equations of motion can be derived as follows:

$$\begin{aligned} EA_u u''_{b0} - K_1 \theta'_b - K_2 \theta'_c + K_3 \psi'_b + K_4 \psi'_c - m_{11} \ddot{u}_{b0} \\ + m_{13} \ddot{\theta}_b + m_{15} \ddot{\theta}_c + f_{bu} = 0 \\ \kappa GA w''_{b0} + K_5 \psi''_b + K_6 \psi''_c - K_7 \theta'_b - K_8 \theta'_c - m_{22} \ddot{w}_{b0} \\ - m_{24} \ddot{\psi}_b - m_{26} \ddot{\psi}_c + f_{bw} = 0 \\ - K_1 u''_{b0} + EI_{\theta_b} \theta''_b + K_9 \theta''_c + K_7 w'_{b0} - K_{10} \psi'_c - GA_b \theta_b \\ + m_{13} \ddot{u}_{b0} - m_{33} \ddot{\theta}_b - m_{35} \ddot{\theta}_c + f_{b\theta} = 0 \\ K_5 w'_{b0} + GI_{\psi_b} \psi'_b + K_{11} \psi''_c - K_3 u'_{b0} - K_{12} \theta'_c \\ - EA_{\psi_b} \psi_b - m_{24} \ddot{w}_{b0} - m_{44} \ddot{\psi}_b - m_{46} \ddot{\psi}_c = 0 \\ - K_2 u''_{b0} + K_9 \theta'_b + EI_{\theta_c} \theta''_c + K_8 w'_{b0} + K_{12} \psi'_b + K_{13} \psi'_c \\ - GA_c \theta_c + m_{15} \ddot{u}_{b0} - m_{35} \ddot{\theta}_b - m_{55} \ddot{\theta}_c = 0 \\ K_6 w'_{b0} + K_{11} \psi''_b + GI_{\psi_c} \psi'_c - K_4 u'_{b0} + K_{10} \theta'_b - K_{13} \theta'_c \\ - EA_{\psi_c} \psi_c - m_{26} \ddot{w}_{b0} - m_{46} \ddot{\psi}_b + m_{66} \ddot{\psi}_c = 0 \end{aligned} \quad (9)$$

where

$$\begin{aligned} EA_u &= A_b C_{11}^* + b A_{11}, & \kappa GA &= \kappa_b Q_{55}^* A_b + \kappa_c b \bar{A}_{55} \\ EI_{\theta_b} &= I_{ob} Q_{11}^* + \frac{1}{4} h_b^2 b A_{11}, & GA_b &= \kappa_b Q_{55}^* A_b \\ GI_{\psi_b} &= \kappa_b C_{55}^* I_{ob} + \frac{1}{4} h_b^2 \kappa_c b \bar{A}_{55}, & EA_{\psi_b} &= A_b C_{33}^* \\ EI_{\theta_c} &= b \bar{D}_{11} + \frac{1}{4} h_c^2 b A_{11}, & GI_{\psi_c} &= \kappa_c b D_{55} + \frac{1}{4} h_c^2 \kappa_c b \bar{A}_{55} \\ GA_c &= \kappa_c b \bar{A}_{55}, & EA_{\psi_c} &= b A_{33} \\ K_1 &= \frac{1}{2} h_b b A_{11}, & K_2 &= \frac{1}{2} h_c b A_{11} \\ K_3 &= A_b C_{13}^*, & K_4 &= b A_{13} \\ K_5 &= \frac{1}{2} h_b \kappa_c b \bar{A}_{55}, & K_6 &= \frac{1}{2} h_c \kappa_c b \bar{A}_{55} \\ K_7 &= \kappa_b Q_{55}^* A_b, & K_8 &= \kappa_c b \bar{A}_{55} \\ K_9 &= \frac{1}{4} h_b h_c b A_{11}, & K_{10} &= \frac{1}{2} h_b b A_{13} \\ K_{11} &= \frac{1}{4} h_c h_b \kappa_c b \bar{A}_{55}, & K_{12} &= \frac{1}{2} h_b \kappa_c b \bar{A}_{55} \\ K_{13} &= \frac{1}{2} h_c \kappa_c b \bar{A}_{55}, & K_{14} &= \frac{1}{2} h_c b A_{13} \end{aligned} \quad (10)$$

and

$$\begin{aligned} m_{11} &= m_{22} = \rho A_c + \rho A_b, & m_{13} &= m_{24} = \frac{1}{2} h_b \rho A_c \\ m_{15} &= \rho R_c + \frac{1}{2} h_c \rho A_c, & m_{26} &= \frac{1}{2} h_c \rho A_c + \rho R_c \\ m_{35} &= m_{46} = \frac{1}{4} h_b h_c \rho A_c + \frac{1}{2} h_b \rho R_c, & m_{33} &= m_{44} = \rho I_{ob} + \frac{1}{4} h_b^2 \rho A_c \\ m_{55} &= \frac{1}{2} h_c \rho R_c + \rho I_{oc} + \frac{1}{4} h_c^2 \rho A_c + \frac{1}{2} h_c \rho R_c \\ m_{66} &= \frac{1}{2} h_c \rho R_c - \rho I_{oc} - \frac{1}{4} h_c^2 \rho A_c - \frac{1}{2} h_c \rho R_c \end{aligned} \quad (11)$$

The natural and geometric boundary conditions, which are consistent with Eq. (9), are also derived from Hamilton's principle as follows:

$$\begin{aligned} N_b(0, t) + N_c(0, t) &= -N_{b1} - N_{c1} & \text{or } \delta u_{b0}(0, t) &= 0 \\ N_b(L, t) + N_c(L, t) &= +N_{b2} + N_{c2} & \text{or } \delta u_{b0}(L, t) &= 0 \\ Q_b(0, t) + Q_c(0, t) &= -Q_{b1} - Q_{c1} & \text{or } \delta w_{b0}(0, t) &= 0 \\ Q_b(L, t) + Q_c(L, t) &= +Q_{b2} + Q_{c2} & \text{or } \delta w_{b0}(L, t) &= 0 \\ M_b(0, t) - \frac{1}{2} h_b N_c(0, t) &= -M_{b1} + \frac{1}{2} h_b N_{c1} & \text{or } \delta \theta_b(0, t) &= 0 \\ M_b(L, t) - \frac{1}{2} h_b N_c(L, t) &= +M_{b2} - \frac{1}{2} h_b N_{c2} & \text{or } \delta \theta_b(L, t) &= 0 \\ R_b(0, t) + \frac{1}{2} h_b Q_c(0, t) &= -\frac{1}{2} h_b Q_{c1} & \text{or } \delta \psi_b(0, t) &= 0 \\ R_b(L, t) + \frac{1}{2} h_b Q_c(L, t) &= +\frac{1}{2} h_b Q_{c2} & \text{or } \delta \psi_b(L, t) &= 0 \\ M_c(0, t) - \frac{1}{2} h_c N_b(0, t) &= +\frac{1}{2} h_c N_{b1} - M_{c1} & \text{or } \delta \theta_c(0, t) &= 0 \\ M_c(L, t) - \frac{1}{2} h_c N_b(L, t) &= -\frac{1}{2} h_c N_{b2} + M_{c2} & \text{or } \delta \theta_c(L, t) &= 0 \\ R_c(0, t) + \frac{1}{2} h_c Q_b(0, t) &= -\frac{1}{2} h_c Q_{b1} & \text{or } \delta \psi_c(0, t) &= 0 \\ R_c(L, t) + \frac{1}{2} h_c Q_b(L, t) &= +\frac{1}{2} h_c Q_{b2} & \text{or } \delta \psi_c(L, t) &= 0 \end{aligned} \quad (12)$$

where $N_b(x, t)$, $Q_b(x, t)$, $M_b(x, t)$, $R_b(x, t)$, $N_c(x, t)$, $Q_c(x, t)$, $M_c(x, t)$, and $R_c(x, t)$ are the resultant forces and moments defined by the force-displacement relations given in [Appendices A and B](#).

3. Spectral element model for a beam with a bonded composite patch

3.1. Governing equations in the frequency domain

The equations of motion (Eq. (9)) can be transformed into the frequency domain by using discrete Fourier transform (DFT) theory [\[26\]](#). To that end, all dependent variables and external forces are represented in spectral forms as

$$\{u_{b0}(x, t), w_{b0}(x, t), \theta_b(x, t), \psi_b(x, t), \theta_c(x, t), \psi_c(x, t)\} \\ = \frac{1}{N} \sum_{n=0}^{N-1} \{U_{bn}(x) W_{bn}(x) \Theta_{bn}(x) \Psi_{bn}(x) \Theta_{cn}(x) \Psi_{cn}(x)\} e^{i\omega_n t} \quad (13)$$

$$\{f_{bu}(x, t), f_{bw}(x, t), f_{b\theta}(x, t)\} = \frac{1}{N} \sum_{n=0}^{N-1} \{F_{un}(x) F_{wn}(x) F_{\theta n}(x)\} e^{i\omega_n t} \quad (14)$$

where $i = \sqrt{-1}$ is the imaginary unit, $\{U_{bn}, W_{bn}, \Theta_{bn}, \Psi_{bn}, \Theta_{cn}, \Psi_{cn}\}$ are the spectral (Fourier) components of the dependent variables, and $\{F_{un}, F_{wn}, F_{\theta n}\}$ are those of the external forces. $N/2$ is the number of spectral components up to the Nyquist frequency [\[26\]](#), and $\omega_n = 2\pi n/T$ are discrete frequencies where T is the sampling time chosen for the fast Fourier transform (FFT)-based spectral analysis.

Substituting Eqs. (13) and (14) into Eq. (9) gives the governing equations in the frequency domain as follows:

$$\begin{aligned}
& EA_u U_b'' - K_1 \Theta_b'' - K_2 \Theta_c'' + K_3 \Psi_b' + K_4 \Psi_c' \\
& + \omega^2 (m_{11} U_b - m_{13} \Theta_b - m_{15} \Theta_c) + F_u = 0 \\
& \kappa GAW_b'' + K_5 \Psi_b'' + K_6 \Psi_c'' - K_7 \Theta_b' - K_8 \Theta_c' \\
& + \omega^2 (m_{22} W_b + m_{24} \Psi_b + m_{26} \Psi_c) + F_w = 0 \\
& - K_1 U_b'' + EI_{\theta_b} \Theta_b'' + K_9 \Theta_c'' + K_7 W_b' - K_{10} \Psi_c' \\
& - GA_b \Theta_b + \omega^2 (-m_{13} U_b + m_{33} \Theta_b + m_{35} \Theta_c) + F_\theta = 0 \\
& K_5 W_b'' + GI_{\psi_b} \Psi_b'' + K_{11} \Psi_c'' - K_3 U_b' - K_{12} \Theta_c' \\
& - EA_{\psi_b} \Psi_b + \omega^2 (m_{24} W_b + m_{44} \Psi_b + m_{46} \Psi_c) = 0 \\
& - K_2 U_b'' + K_9 \Theta_b'' + EI_{\theta_c} \Theta_c'' + K_8 W_b' + K_{12} \Psi_b' + K_{13} \Psi_c' \\
& - GA_c \Theta_c + \omega^2 (-m_{15} U_b + m_{35} \Theta_b + m_{55} \Theta_c) = 0 \\
& K_6 W_b'' + K_{11} \Psi_b'' + GI_{\psi_c} \Psi_c'' - K_4 U_b' + K_{10} \Theta_b' - K_{13} \Theta_c' \\
& - EA_{\psi_c} \Psi_c + \omega^2 (m_{26} W_b + m_{46} \Psi_b - m_{66} \Psi_c) = 0
\end{aligned} \quad (15)$$

where the subscripts n are omitted for the sake of brevity. Similarly, by representing all resultant forces and moments, boundary forces and moments, and the externally applied voltage in spectral forms, the natural boundary conditions given by Eq. (12) and the force-displacement relations given in Appendices A and B can be transformed into the frequency domain as follows:

$$\begin{aligned}
\bar{N}_b(0) + \bar{N}_c(0) &= -\bar{N}_{b1} - \bar{N}_{c1}, & \bar{N}_b(L) + N_c(L) &= \bar{N}_{b2} + \bar{N}_{c2} \\
\bar{Q}_b(0) + \bar{Q}_c(0) &= -\bar{Q}_{b1} - \bar{Q}_{c1}, & \bar{Q}_b(L) + \bar{Q}_c(L) &= \bar{Q}_{b2} + \bar{Q}_{c2} \\
\bar{M}_b(0) + \frac{1}{2} h_b \bar{N}_c(0) &= -\bar{M}_{b1} + \frac{1}{2} h_b \bar{N}_{c1}, & \bar{M}_b(L) + \frac{1}{2} h_b \bar{N}_c(L) &= \bar{M}_{b2} - \frac{1}{2} h_b \bar{N}_{c2} \\
\bar{R}_b(0) + \frac{1}{2} h_b \bar{Q}_c(0) &= -\frac{1}{2} h_b \bar{Q}_{c1}, & \bar{R}_b(L) + \frac{1}{2} h_b \bar{Q}_c(L) &= \frac{1}{2} h_b \bar{Q}_{c2} \\
\bar{M}_c(0) - \frac{1}{2} h_c \bar{N}_c(0) &= -\bar{M}_{c1} + \frac{1}{2} h_c \bar{N}_{c1}, & \bar{M}_c(L) - \frac{1}{2} h_c \bar{N}_c(L) &= \bar{M}_{c2} - \frac{1}{2} h_c \bar{N}_{c2} \\
\bar{R}_c(0) + \frac{1}{2} h_c \bar{Q}_c(0) &= -\frac{1}{2} h_c \bar{Q}_{c1}, & \bar{R}_c(L) + \frac{1}{2} h_c \bar{Q}_c(L) &= \frac{1}{2} h_c \bar{Q}_{c2}
\end{aligned} \quad (16)$$

and

$$\begin{aligned}
\bar{N}_b(x) &= A_b C_{11}^* U' + A_b C_{13}^* \Psi_b \\
\bar{Q}_b(x) &= \kappa_b C_{55}^* A_b (W' - \Theta_b) \\
\bar{M}_b(x) &= I_{ob} C_{11}^* \Theta_b' \\
\bar{R}_b(x) &= \kappa_b C_{55}^* I_{ob} \Psi_b' \\
\bar{N}_c(x) &= bA_{11} \left(U' - \frac{1}{2} h_b \Theta_b' - \frac{1}{2} h_c \Theta_c' \right) - bB_{11} \Theta_c' + bA_{13} \Psi_c \\
\bar{Q}_c(x) &= \kappa_c bA_{55} \left(W' + \frac{1}{2} h_b \Psi_b' + \frac{1}{2} h_c \Psi_c' - \Theta_c \right) + \kappa_c bB_{55} \Psi_c' \\
\bar{M}_c(x) &= -bB_{11} \left(U' - \frac{1}{2} h_b \Theta_b' - \frac{1}{2} h_c \Theta_c' \right) + bD_{11} \Theta_c' - bB_{13} \Psi_c \\
\bar{R}_c(x) &= \kappa_c bB_{55} \left(W' + \frac{1}{2} h_b \Psi_b' + \frac{1}{2} h_c \Psi_c' - \Theta_c \right) + \kappa_c bD_{55} \Psi_c'
\end{aligned} \quad (17)$$

3.2. Dispersion relation and free waves

The free wave solutions in the frequency domain can be obtained from homogeneous governing equations that can be reduced from Eq. (15) by removing all external forces. Thus,

$$\begin{aligned}
& EA_u U_b'' - K_1 \Theta_b'' - K_2 \Theta_c'' + K_3 \Psi_b' + K_4 \Psi_c' \\
& + \omega^2 (m_{11} U_b - m_{13} \Theta_b - m_{15} \Theta_c) = 0 \\
& \kappa GAW_b'' + K_5 \Psi_b'' + K_6 \Psi_c'' - K_7 \Theta_b' - K_8 \Theta_c' \\
& + \omega^2 (m_{22} W_b + m_{24} \Psi_b + m_{26} \Psi_c) = 0 \\
& - K_1 U_b'' + EI_{\theta_b} \Theta_b'' + K_9 \Theta_c'' + K_7 W_b' - K_{10} \Psi_c' \\
& - GA_b \Theta_b + \omega^2 (-m_{13} U_b + m_{33} \Theta_b + m_{35} \Theta_c) = 0 \\
& K_5 W_b'' + GI_{\psi_b} \Psi_b'' + K_{11} \Psi_c'' - K_3 U_b' - K_{12} \Theta_c' - EA_{\psi_b} \Psi_b \\
& + \omega^2 (m_{24} W_b + m_{44} \Psi_b + m_{46} \Psi_c) = 0
\end{aligned}$$

$$\begin{aligned}
& - K_2 U_b'' + K_9 \Theta_b'' + EI_{\theta_c} \Theta_c'' + K_8 W_b' + K_{12} \Psi_b' + (K_{13} - K_{14}) \Psi_c' \\
& - GA_c \Theta_c + \omega^2 (-m_{15} U_b + m_{35} \Theta_b + m_{55} \Theta_c) = 0 \\
& K_6 W_b'' + K_{11} \Psi_b'' + GI_{\psi_c} \Psi_c'' - K_4 U_b' + K_{10} \Theta_b' - (K_{13} - K_{14}) \Theta_c' \\
& - EA_{\psi_c} \Psi_c + \omega^2 (m_{26} W_b + m_{46} \Psi_b - m_{66} \Psi_c) = 0
\end{aligned} \quad (18)$$

We assumed the solutions of Eq. (18) to be of the form

$$\begin{aligned}
U_b(x) &= ae^{-ikx}, & W_b(x) &= r_1 ae^{-ikx}, & \Theta_b(x) &= r_2 ae^{-ikx} \\
\Psi_b(x) &= r_3 ae^{-ikx}, & \Theta_c(x) &= r_4 ae^{-ikx}, & \Psi_c(x) &= r_5 ae^{-ikx}
\end{aligned} \quad (19)$$

where a is a constant. Then, substituting Eq. (19) into Eq. (18) yields the following eigenvalue problem:

$$\begin{bmatrix} X_{11} & 0 & X_{13} & X_{14} & X_{15} & X_{16} \\ 0 & X_{22} & X_{23} & X_{24} & X_{25} & X_{26} \\ X_{13} & -X_{23} & X_{33} & 0 & X_{35} & X_{36} \\ -X_{14} & X_{24} & 0 & X_{44} & X_{45} & X_{46} \\ X_{15} & -X_{25} & X_{35} & -X_{45} & X_{55} & X_{56} \\ -X_{16} & X_{26} & -X_{36} & X_{46} & -X_{56} & X_{66} \end{bmatrix} \begin{Bmatrix} r_1 \\ r_2 \\ r_3 \\ r_4 \\ r_5 \end{Bmatrix} = 0 \quad (20)$$

where

$$\begin{aligned}
X_{11} &= -EA_u k^2 + \omega^2 m_{11}, & X_{13} &= K_1 k^2 - \omega^2 m_{13} \\
X_{14} &= -iK_3 k, & X_{15} &= K_2 k^2 - \omega^2 m_{15} \\
X_{16} &= -iK_4 k, & X_{22} &= -\kappa GAK^2 + \omega^2 m_{22} \\
X_{23} &= iK_7 k, & X_{24} &= -K_5 k^2 + \omega^2 m_{24}, \\
X_{25} &= iK_8 k, & X_{26} &= -K_6 k^2 + \omega^2 m_{26} \\
X_{33} &= -EI_{\theta_b} k^2 - GA_b + \omega^2 m_{33}, & X_{35} &= -K_9 k^2 + \omega^2 m_{35} \\
X_{36} &= iK_{10} k, & X_{44} &= -GI_{\psi_b} k^2 - EA_{\psi_b} + \omega^2 m_{44} \\
X_{45} &= iK_{12} k, & X_{46} &= -K_{11} k^2 + \omega^2 m_{46} \\
X_{55} &= -EI_{\theta_c} k^2 - GA_c + \omega^2 m_{55}, & X_{56} &= -i(K_{13} - K_{14})k \\
X_{66} &= -GI_{\psi_c} k^2 - EA_{\psi_c} - \omega^2 m_{66}
\end{aligned} \quad (21)$$

From the condition for the existence of nontrivial solutions of the eigenvalue problem, we can derive a dispersion relation from Eq. (20) as follows:

$$a_1 \lambda^6 + a_2 \lambda^5 + a_3 \lambda^4 + a_4 \lambda^3 + a_5 \lambda^2 + a_6 \lambda + a_7 = 0 \quad (\lambda = k^2) \quad (22)$$

from which twelve wavenumbers can be computed:

$$\begin{aligned}
k_1 &= -k_7 = \sqrt{\lambda_1}, & k_2 &= -k_8 = \sqrt{\lambda_2}, & k_3 &= -k_9 = \sqrt{\lambda_3} \\
k_4 &= -k_{10} = \sqrt{\lambda_4}, & k_5 &= -k_{11} = \sqrt{\lambda_5}, & k_6 &= -k_{12} = \sqrt{\lambda_6}
\end{aligned} \quad (23)$$

For each wavenumber $k_q (q = 1, 2, \dots, 12)$, the coefficients $r_p (p = 1, 2, 3, 4, 5)$ in Eq. (19) can be then determined from Eq. (20) as follows:

$$\begin{Bmatrix} r_1 \\ r_2 \\ r_3 \\ r_4 \\ r_5 \end{Bmatrix}_{k=k_q} \equiv \begin{Bmatrix} r_{1q} \\ r_{2q} \\ r_{3q} \\ r_{4q} \\ r_{5q} \end{Bmatrix} = - \begin{bmatrix} X_{22} & X_{23} & X_{24} & X_{25} & X_{26} \\ -X_{23} & X_{33} & 0 & X_{35} & X_{36} \\ X_{24} & 0 & X_{44} & X_{45} & X_{46} \\ -X_{25} & X_{35} & -X_{45} & X_{55} & X_{56} \\ X_{26} & -X_{36} & X_{46} & -X_{56} & X_{66} \end{bmatrix}^{-1} \times \begin{Bmatrix} 0 \\ X_{13} \\ -X_{14} \\ X_{15} \\ -X_{16} \end{Bmatrix}_{k=k_q} \quad (24)$$

By applying Eqs. (23) and (24) to Eq. (19), the general solutions of Eq. (18) can be written as follows:

$$\begin{aligned}
U_b(x) &= \sum_{q=1}^{12} a_q e^{-ik_q x} = \mathbf{e}(x, \omega) \mathbf{a}, & W_b(x) &= \sum_{q=1}^{12} r_{1q} a_q e^{-ik_q x} = \mathbf{e}(x, \omega) \mathbf{R}_1 \mathbf{a} \\
\Theta_b(x) &= \sum_{q=1}^{12} r_{2q} a_q e^{-ik_q x} = \mathbf{e}(x, \omega) \mathbf{R}_2 \mathbf{a}, & \Psi_b(x) &= \sum_{q=1}^{12} r_{3q} a_q e^{-ik_q x} = \mathbf{e}(x, \omega) \mathbf{R}_3 \mathbf{a} \\
\Theta_c(x) &= \sum_{q=1}^{12} r_{4q} a_q e^{-ik_q x} = \mathbf{e}(x, \omega) \mathbf{R}_4 \mathbf{a}, & \Psi_c(x) &= \sum_{q=1}^{12} r_{5q} a_q e^{-ik_q x} = \mathbf{e}(x, \omega) \mathbf{R}_5 \mathbf{a}
\end{aligned} \quad (25)$$

where

$$\mathbf{e}(x, \omega) = [e^{-ik_1 x} \quad e^{-ik_2 x} \quad e^{-ik_3 x} \quad \dots \quad e^{-ik_{12} x}] \quad (26)$$

and

$$\begin{aligned}
\mathbf{a} &= \{a_1 \ a_2 \ \dots \ a_{12}\}^T \\
\mathbf{R}_p(\omega) &= \text{diag}[r_{pq}(\omega)] \quad (p = 1, 2, 3, 4, 5; q = 1, 2, 3, \dots, 12)
\end{aligned} \quad (27)$$

Notice that the general solutions in Eq. (25) are now expressed in terms of free waves that propagate forward and backward in a composite patched beam.

3.3. Spectral nodal DOFs and dynamic shape functions

Consider a finite composite patched beam element of length l as shown in Fig. 2. The nodal degrees of freedom (DOFs) in the frequency domain (called spectral nodal DOFs) are defined as

$$\begin{aligned}
U_{b1} &= U_b(0), & U_{b2} &= U_b(l) \\
W_{b1} &= W_b(0), & W_{b2} &= W_b(l) \\
\Theta_{b1} &= \Theta_b(0), & \Theta_{b2} &= \Theta_b(l) \\
\Psi_{b1} &= \Psi_b(0), & \Psi_{b2} &= \Psi_b(l) \\
\Theta_{c1} &= \Theta_c(0), & \Theta_{c2} &= \Theta_c(l) \\
\Psi_{c1} &= \Psi_c(0), & \Psi_{c2} &= \Psi_c(l)
\end{aligned} \quad (28)$$

By using Eq. (25), the spectral nodal DOFs can be expressed in terms of constant vector $\mathbf{a}$ as

$$\mathbf{d}_{BC} = \mathbf{H}(\omega) \mathbf{a} \quad (29)$$

where

$$\mathbf{d}_{BC} = \{U_{b1} \ W_{b1} \ \Theta_{b1} \ \Psi_{b1} \ \Theta_{c1} \ \Psi_{c1} \ U_{b2} \ W_{b2} \ \Theta_{b2} \ \Psi_{b2} \ \Theta_{c2} \ \Psi_{c2}\}^T \quad (30)$$

and

$$\mathbf{H}(\omega) = \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ r_{11} & r_{12} & r_{13} & \dots & r_{112} \\ r_{21} & r_{22} & r_{23} & \dots & r_{212} \\ r_{31} & r_{32} & r_{33} & \dots & r_{312} \\ r_{41} & r_{42} & r_{43} & \dots & r_{412} \\ r_{51} & r_{52} & r_{53} & \dots & r_{512} \\ e_1 & e_2 & e_3 & \dots & e_{12} \\ r_{11}e_1 & r_{12}e_2 & r_{13}e_3 & \dots & r_{112}e_{12} \\ r_{21}e_1 & r_{22}e_2 & r_{23}e_3 & \dots & r_{212}e_{12} \\ r_{31}e_1 & r_{32}e_2 & r_{33}e_3 & \dots & r_{312}e_{12} \\ r_{41}e_1 & r_{42}e_2 & r_{43}e_3 & \dots & r_{412}e_{12} \\ r_{51}e_1 & r_{52}e_2 & r_{53}e_3 & \dots & r_{512}e_{12} \end{bmatrix} \quad (31)$$

where

$$e_q = e^{-ik_q l} \quad (q = 1, 2, \dots, 12) \quad (32)$$

By using Eq. (29), constant vector $\mathbf{a}$ can be eliminated from Eq. (25) to write the general solutions in terms of the nodal DOFs vector $\mathbf{d}_{BC}$ as

$$\begin{aligned}
U_b(x) &= \mathbf{N}_u(x; \omega) \mathbf{d}_{BC}, & W_b(x) &= \mathbf{N}_w(x; \omega) \mathbf{d}_{BC} \\
\Theta_b(x) &= \mathbf{N}_{\theta_b}(x; \omega) \mathbf{d}_{BC}, & \Psi_b(x) &= \mathbf{N}_{\psi_b}(x; \omega) \mathbf{d}_{BC} \\
\Theta_c(x) &= \mathbf{N}_{\theta_c}(x; \omega) \mathbf{d}_{BC}, & \Psi_c(x) &= \mathbf{N}_{\psi_c}(x; \omega) \mathbf{d}_{BC}
\end{aligned} \quad (33)$$

where $\mathbf{N}_u$, $\mathbf{N}_w$, $\mathbf{N}_{\theta_b}$, $\mathbf{N}_{\psi_b}$, $\mathbf{N}_{\theta_c}$, and $\mathbf{N}_{\psi_c}$ are the frequency-dependent dynamic shape functions, defined by

$$\begin{aligned}
\mathbf{N}_u(x; \omega) &= \mathbf{e}(x; \omega) \mathbf{H}(\omega)^{-1} \\
\mathbf{N}_w(x; \omega) &= \mathbf{e}(x; \omega) \mathbf{R}_1(\omega) \mathbf{H}(\omega)^{-1} \\
\mathbf{N}_{\theta_b}(x; \omega) &= \mathbf{e}(x; \omega) \mathbf{R}_2(\omega) \mathbf{H}(\omega)^{-1} \\
\mathbf{N}_{\psi_b}(x; \omega) &= \mathbf{e}(x; \omega) \mathbf{R}_3(\omega) \mathbf{H}(\omega)^{-1} \\
\mathbf{N}_{\theta_c}(x; \omega) &= \mathbf{e}(x; \omega) \mathbf{R}_4(\omega) \mathbf{H}(\omega)^{-1} \\
\mathbf{N}_{\psi_c}(x; \omega) &= \mathbf{e}(x; \omega) \mathbf{R}_5(\omega) \mathbf{H}(\omega)^{-1}
\end{aligned} \quad (34)$$

3.4. Formulation of spectral element equation

The spectral element model for the composite patched beam can be formulated by using the variational method [18,27] from the weak form of the governing equations in the frequency domain. The weak form of the governing equations can be derived from the weighted integral statement of Eq. (15), given by

$$\begin{aligned}
\int_0^l & [\{EA_u U_b'' - K_1 \Theta_b'' - K_2 \Theta_c'' + K_3 \Psi_b' + K_4 \Psi_c' \\
& + \omega^2(m_{11}U_b - m_{13}\Theta_b - m_{15}\Theta_c) + F_u\} \delta U_b \\
& + \{\kappa GAW_b'' + K_5 \Psi_b'' + K_6 \Psi_c'' - K_7 \Theta_b' - K_8 \Theta_c' \\
& + \omega^2(m_{22}W_b + m_{24}\Psi_b + m_{26}\Psi_c) + F_w\} \delta W_b \\
& + \{-K_1 U_b' + EI_{\theta_b} \Theta_b'' + K_9 \Theta_c'' + K_7 W_b' - K_{10} \Psi_c' - GA_b \Theta_b \\
& + \omega^2(-m_{13}U_b + m_{33}\Theta_b + m_{35}\Theta_c) + F_{\theta}\} \delta \Theta_b \\
& + \{K_5 W_b' + GI_{\psi_b} \Psi_b'' + K_{11} \Psi_c'' - K_3 U_b' - K_{12} \Theta_c' - EA_{\psi_b} \Psi_b \\
& + \omega^2(m_{24}W_b + m_{44}\Psi_b + m_{46}\Psi_c)\} \delta \Psi_b \\
& + \{-K_2 U_b' + K_9 \Theta_b' + EI_{\theta_c} \Theta_c'' + K_8 W_b' + K_{12} \Psi_b' \\
& + (K_{13} - K_{14}) \Psi_c' - GA_c \Theta_c + \omega^2(-m_{15}U_b + m_{35}\Theta_b + m_{55}\Theta_c)\} \delta \Theta_c \\
& + \{K_6 W_b' + K_{11} \Psi_b'' + GI_{\psi_c} \Psi_c'' - K_4 U_b' + K_{10} \Theta_b' - (K_{13} - K_{14}) \Theta_c' \\
& - EA_{\psi_c} \Psi_c + \omega^2(m_{26}W_b + m_{46}\Psi_b - m_{66}\Psi_c)\} \delta \Psi_c] dx = 0 \quad (35)
\end{aligned}$$

Applying integration by parts to Eq. (35) and imposing natural boundary conditions given by Eq. (16), we obtain the weak form as

$$\begin{aligned}
\int_0^l & [(EA_u U_b' - K_1 \Theta_b' - K_2 \Theta_c' + K_3 \Psi_b + K_4 \Psi_c) \delta U_b' \\
& + (\kappa GAW_b' + K_5 \Psi_b' + K_6 \Psi_c' - K_7 \Theta_b - K_8 \Theta_c) \delta W_b' \\
& + (-K_1 U_b' + EI_{\theta_b} \Theta_b' + K_9 \Theta_c' - K_{10} \Psi_c) \delta \Theta_b' + (-K_7 W_b' + GA_b \Theta_b) \delta \Theta_b \\
& + (K_5 W_b' + GI_{\psi_b} \Psi_b' + K_{11} \Psi_c' - K_{12} \Theta_c) \delta \Psi_b' + (K_3 U_b' + EA_{\psi_b} \Psi_b) \delta \Psi_b \\
& + (-K_2 U_b' + K_9 \Theta_b' + EI_{\theta_c} \Theta_c' - K_{14} \Psi_c) \delta \Theta_c' \\
& + (-K_8 W_b' - K_{12} \Psi_b' - K_{13} \Psi_c' + GA_c \Theta_c) \delta \Theta_c \\
& + (K_6 W_b' + K_{11} \Psi_b' + GI_{\psi_c} \Psi_c' - K_{13} \Theta_c) \delta \Psi_c' \\
& + (K_4 U_b' - K_{10} \Theta_b' - K_{14} \Theta_c' + EA_{\psi_c} \Psi_c) \delta \Psi_c \\
& - \omega^2 \{ (m_{11}U_b - m_{13}\Theta_b - m_{15}\Theta_c) \delta U_b \\
& + (m_{22}W_b + m_{24}\Psi_b + m_{26}\Psi_c) \delta W_b \\
& + (-m_{13}U_b + m_{33}\Theta_b + m_{35}\Theta_c) \delta \Theta_b \\
& + (m_{24}W_b + m_{44}\Psi_b + m_{46}\Psi_c) \delta \Psi_b \\
& + (-m_{15}U_b + m_{35}\Theta_b + m_{55}\Theta_c) \delta \Theta_c + m_{26}W_b \\
& + m_{46}\Psi_b - m_{66}\Psi_c \} \delta \Psi_c] dx \\
& = \delta \mathbf{d}^T \mathbf{f} + \int_0^L (F_u \delta U_b + F_w \delta W_b + F_{\theta} \delta \Theta_b) dx \quad (36)
\end{aligned}$$

where

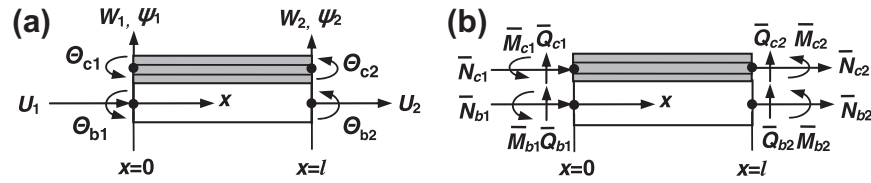


Fig. 2. Finite element of the composite patched beam: (a) spectral nodal DOFs and (b) spectral nodal forces and moments.

$$\mathbf{f}_c = \left\{ \bar{N}_{b1} + \bar{N}_{c1} \bar{Q}_{b1} + \bar{Q}_{c1} \bar{M}_{b1} - \frac{1}{2} h_b \bar{Q}_{c1} \frac{1}{2} h_b \bar{Q}_{c1} M_{c1} \right. \\ \left. - \frac{1}{2} h_c \bar{N}_{c1} \frac{1}{2} h_c \bar{Q}_{c1} \bar{N}_{b2} + \bar{N}_{c2} \bar{Q}_{b2} + \bar{Q}_{c2} \bar{M}_{b2} \right. \\ \left. - \frac{1}{2} h_b \bar{Q}_{c2} \frac{1}{2} h_b \bar{Q}_{c2} \bar{M}_{c2} - \frac{1}{2} h_c \bar{N}_{c2} \frac{1}{2} h_c \bar{Q}_{c2} \right\}^T \quad (37)$$

By substituting Eq. (33) into Eq. (36) and performing some manipulations, we obtain the spectral element equation as follows:

$$\mathbf{S}_{BC}(\omega) \mathbf{d}_{BC} = \mathbf{f}_{BC}(\omega) \quad (38)$$

where $\mathbf{S}_{BC}(\omega)$ is a 12-by-12 frequency-dependent symmetric dynamic stiffness matrix called the spectral element matrix, and $\mathbf{f}_{BC}(\omega)$ is the spectral nodal forces vector. These are given by

$$\mathbf{S}_{BC}(\omega) = \mathbf{H}(\omega)^{-T} \mathbf{D}(\omega) \mathbf{H}(\omega)^{-1} \quad (39)$$

$$\mathbf{f}_{BC}(\omega) = \mathbf{f}_d(\omega) + \mathbf{f}_c(\omega) \quad (40)$$

where

$$\mathbf{D}(\omega) = -EA_u \mathbf{K} \mathbf{E} \mathbf{K} - \kappa G A_r \mathbf{K} \mathbf{E} \mathbf{K} \mathbf{R}_1 - EI_{\theta_b} \mathbf{R}_2 \mathbf{K} \mathbf{E} \mathbf{K} \mathbf{R}_2 \\ - GI_{\psi_b} \mathbf{R}_3 \mathbf{K} \mathbf{E} \mathbf{K} \mathbf{R}_3 - EI_{\theta_c} \mathbf{R}_4 \mathbf{K} \mathbf{E} \mathbf{K} \mathbf{R}_4 - GI_{\psi_c} \mathbf{R}_5 \mathbf{K} \mathbf{E} \mathbf{K} \mathbf{R}_5 \\ + GA_b \mathbf{R}_2 \mathbf{E} \mathbf{R}_2 + EA_{\psi_b} \mathbf{R}_3 \mathbf{E} \mathbf{R}_3 + GA_c \mathbf{R}_4 \mathbf{E} \mathbf{R}_4 + EA_{\psi_c} \mathbf{R}_5 \mathbf{E} \mathbf{R}_5 \\ + K_1 (\mathbf{K} \mathbf{E} \mathbf{K} \mathbf{R}_2 + \mathbf{R}_2 \mathbf{K} \mathbf{E} \mathbf{K}) + K_2 (\mathbf{K} \mathbf{E} \mathbf{K} \mathbf{R}_4 + \mathbf{R}_4 \mathbf{K} \mathbf{E} \mathbf{K}) \\ - iK_3 (\mathbf{K} \mathbf{E} \mathbf{R}_3 + \mathbf{R}_3 \mathbf{E} \mathbf{K}) - iK_4 (\mathbf{K} \mathbf{E} \mathbf{R}_5 + \mathbf{R}_5 \mathbf{E} \mathbf{K}) \\ - K_5 (\mathbf{R}_1 \mathbf{K} \mathbf{E} \mathbf{K} \mathbf{R}_3 + \mathbf{R}_3 \mathbf{K} \mathbf{E} \mathbf{K} \mathbf{R}_1) - K_6 (\mathbf{R}_1 \mathbf{K} \mathbf{E} \mathbf{K} \mathbf{R}_5 \\ + \mathbf{R}_5 \mathbf{K} \mathbf{E} \mathbf{K} \mathbf{R}_1) + iK_7 (\mathbf{R}_1 \mathbf{K} \mathbf{E} \mathbf{R}_2 + \mathbf{R}_2 \mathbf{E} \mathbf{K} \mathbf{R}_1) \\ + iK_8 (\mathbf{R}_1 \mathbf{K} \mathbf{E} \mathbf{R}_4 + \mathbf{R}_4 \mathbf{E} \mathbf{K} \mathbf{R}_1) - K_9 (\mathbf{R}_2 \mathbf{K} \mathbf{E} \mathbf{K} \mathbf{R}_4 \\ + \mathbf{R}_4 \mathbf{K} \mathbf{E} \mathbf{K} \mathbf{R}_2) + iK_{10} (\mathbf{R}_2 \mathbf{K} \mathbf{E} \mathbf{R}_5 + \mathbf{R}_5 \mathbf{E} \mathbf{K} \mathbf{R}_2) \\ - K_{11} (\mathbf{R}_3 \mathbf{K} \mathbf{E} \mathbf{K} \mathbf{R}_5 + \mathbf{R}_5 \mathbf{K} \mathbf{E} \mathbf{K} \mathbf{R}_3) + iK_{12} (\mathbf{R}_3 \mathbf{K} \mathbf{E} \mathbf{R}_4 \\ + \mathbf{R}_4 \mathbf{E} \mathbf{K} \mathbf{R}_3) + iK_{13} (\mathbf{R}_4 \mathbf{E} \mathbf{K} \mathbf{R}_5 + \mathbf{R}_5 \mathbf{E} \mathbf{K} \mathbf{R}_4) \\ + iK_{14} (\mathbf{R}_4 \mathbf{K} \mathbf{E} \mathbf{R}_5 + \mathbf{R}_5 \mathbf{E} \mathbf{K} \mathbf{R}_4) - \omega^2 [m_{11} \mathbf{E} + m_{22} \mathbf{R}_1 \mathbf{E} \mathbf{R}_1 \\ + m_{33} \mathbf{R}_2 \mathbf{E} \mathbf{R}_2 + m_{44} \mathbf{R}_3 \mathbf{E} \mathbf{R}_3 + m_{55} \mathbf{R}_4 \mathbf{E} \mathbf{R}_4 - m_{66} \mathbf{R}_5 \mathbf{E} \mathbf{R}_5 \\ - m_{13} (\mathbf{E} \mathbf{R}_2 + \mathbf{R}_2 \mathbf{E}) - m_{15} (\mathbf{E} \mathbf{R}_4 + \mathbf{R}_4 \mathbf{E}) + m_{24} (\mathbf{R}_1 \mathbf{E} \mathbf{R}_3 \\ + \mathbf{R}_3 \mathbf{E} \mathbf{R}_1) + m_{26} (\mathbf{R}_1 \mathbf{E} \mathbf{R}_5 + \mathbf{R}_5 \mathbf{E} \mathbf{R}_1) + m_{35} (\mathbf{R}_2 \mathbf{E} \mathbf{R}_4 \\ + \mathbf{R}_4 \mathbf{E} \mathbf{R}_2) + m_{46} (\mathbf{R}_3 \mathbf{E} \mathbf{R}_5 + \mathbf{R}_5 \mathbf{E} \mathbf{R}_3)] \quad (41)$$

$$\mathbf{f}_d(\omega) = \int_0^l \left(\mathbf{N}_u^T F_u + \mathbf{N}_w^T F_w + \mathbf{N}_{\theta_b}^T F_{\theta} \right) dx \quad (42)$$

with the following definitions:

$$\mathbf{K}(\omega) = \text{diag}[k_q] \quad (q = 1, 2, \dots, 12) \quad (43)$$

$$\mathbf{E}(\omega) = \int_0^l \mathbf{e}(x; \omega)^T \mathbf{e}(x; \omega) dx = [E_{pq}(\omega)] \quad (p, q = 1, 2, \dots, 12) \quad (44)$$

where

$$E_{pq}(\omega) = \begin{cases} \frac{i}{k_p + k_q} [e^{-i(k_p + k_q)l} - 1] & \text{if } k_p + k_q \neq 0 \\ l & \text{if } k_p + k_q = 0 \end{cases} \quad (45)$$

4. Spectral element model for the beam with a debonded composite patch

There are three types of debonding: (a) right-edge debonding, (b) inside debonding, and (c) left-edge debonding. The spectral element modeling procedure for right-edge debonding is the same as that for left-edge debonding. The spectral element model for inside debonding can be obtained by assembling the spectral element models of the right-edge and left-edge debondings. Thus, the spectral element modeling procedure will be discussed herein only for the case of right-edge debonding.

A beam element with a right-edge debonded composite patch is shown in Fig. 3, and can be considered as an assembly of three parts: a composite patched beam element, a (metallic) beam element, and a composite patch element. The spectral element equation for the composite patched beam element is given by Eq. (38) and can be written in partitioned form as

$$\mathbf{S}_{BC}(\omega) \mathbf{d}_{BC} = \mathbf{f}_{BC}, \quad \mathbf{S}_{BC}(\omega) = \begin{bmatrix} \mathbf{S}_{BC}^{11} & \mathbf{S}_{BC}^{12} \\ \mathbf{S}_{BC}^{21} & \mathbf{S}_{BC}^{22} \end{bmatrix}_{(12 \times 12)} \quad (46)$$

The spectral element equations for the (metallic) beam element and the composite patch element can be obtained in the following forms (without providing details for the sake of brevity):

$$\mathbf{S}_B(\omega) \mathbf{d}_B = \mathbf{f}_B, \quad \mathbf{S}_B(\omega) = \begin{bmatrix} \mathbf{S}_B^{11} & \mathbf{S}_B^{12} \\ \mathbf{S}_B^{21} & \mathbf{S}_B^{22} \end{bmatrix}_{(8 \times 8)} \quad (\text{beam element}) \quad (47)$$

$$\mathbf{S}_C(\omega) \mathbf{d}_C = \mathbf{f}_C, \quad \mathbf{S}_C(\omega) = \begin{bmatrix} \mathbf{S}_C^{11} & \mathbf{S}_C^{12} \\ \mathbf{S}_C^{21} & \mathbf{S}_C^{22} \end{bmatrix}_{(8 \times 8)} \quad (\text{composite patch element}) \quad (48)$$

where $\mathbf{S}_B$ and $\mathbf{S}_C$ are 8-by-8 symmetric spectral element matrices, and $\mathbf{d}_B$ and $\mathbf{d}_C$ are defined by

$$\mathbf{d}_B = \{U_{b5} \ W_{b5} \ \Theta_{b5} \ \Psi_{b5} \ U_{b6} \ W_{b6} \ \Theta_{b6} \ \Psi_{b6}\}^T \quad (49)$$

$$\mathbf{d}_C = \{U_{c3} \ W_{c3} \ \Theta_{c3} \ \Psi_{c3} \ U_{c4} \ W_{c4} \ \Theta_{c4} \ \Psi_{c4}\}^T$$

The spectral nodal DOFs at node 3 of the composite patch element and at node 5 of the beam element can all be related to those at node 2 of the composite patched beam element as follows:

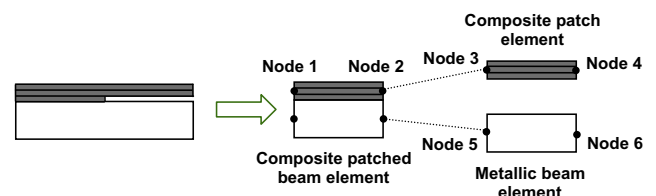


Fig. 3. Beam element with a right-edge debonded composite patch.

$$\begin{aligned}
U_{c3} &= U_{c2} = U_{b2} - \frac{1}{2}h_b\theta_{b2} - \frac{1}{2}h_c\theta_{c2} \\
W_{c3} &= W_{c2} = W_{b2} + \frac{1}{2}h_b\psi_{b2} + \frac{1}{2}h_c\psi_{c2} \\
\theta_{c3} &= \theta_{c2}, \quad \psi_{c3} = \psi_{c2} \\
U_{b5} &= U_{b2}, \quad W_{b5} = W_{b2} \\
\theta_{b5} &= \theta_{b2}, \quad \psi_{b5} = \psi_{b2}
\end{aligned} \quad (50)$$

By using Eq. (50), three spectral element equations (Eqs. (46)–(48)) can be assembled in the following form:

$$\mathbf{S}^R(\omega)\mathbf{d}^R = \mathbf{f}^R \quad (51)$$

where $\mathbf{S}^R(\omega)$ is the 20-by-20 spectral element matrix, $\mathbf{d}^R$ is the 20-by-1 nodal DOFs vector, and $\mathbf{f}^R$ is the 20-by-1 spectral nodal forces vector: they are defined by

$$\mathbf{S}^R(\omega) = \begin{bmatrix} \mathbf{S}_{BC}^{11} & \mathbf{S}_{BC}^{12} & 0 & 0 \\ \mathbf{S}_{BC}^{21} & \mathbf{S}_{BC}^{22} + \mathbf{T}_1^T \mathbf{S}_C^{11} \mathbf{T}_1 + \mathbf{T}_2^T \mathbf{S}_B^{11} \mathbf{T}_2 & \mathbf{T}_1^T \mathbf{S}_C^{12} & \mathbf{T}_2^T \mathbf{S}_B^{12} \\ 0 & \mathbf{S}_C^{21} \mathbf{T}_1 & \mathbf{S}_C^{22} & 0 \\ 0 & \mathbf{S}_B^{21} \mathbf{T}_2 & 0 & \mathbf{S}_B^{22} \end{bmatrix}$$

$$\mathbf{d}^R = \{U_{b1} \ W_{b1} \ \theta_{b1} \ \psi_{b1} \ \theta_{c1} \ \psi_{c1} \ U_{b2} \ W_{b2} \ \theta_{b2} \ \psi_{b2} \ \theta_{c2} \ \psi_{c2} \ U_{c4} \ W_{c4} \ \theta_{c4} \ \psi_{c4} \ U_{b6} \ W_{b6} \ \theta_{b6} \ \psi_{b6}\}^T \quad (52)$$

$$\mathbf{f}^R = \{\mathbf{f}_{BC}^1 \ \mathbf{f}_{BC}^2 + \mathbf{T}_1^T \mathbf{f}_C^1 + \mathbf{T}_2^T \mathbf{f}_B^1 \ \mathbf{f}_C^2 \ \mathbf{f}_B^2\}^T$$

where

$$\mathbf{T}_1 = \begin{bmatrix} 1 & 0 & -\frac{1}{2}h_b & 0 & -\frac{1}{2}h_c & 0 \\ 0 & 1 & 0 & \frac{1}{2}h_b & 0 & \frac{1}{2}h_c \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{T}_2 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix} \quad (53)$$

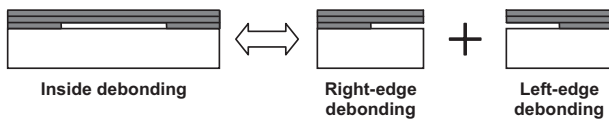


Fig. 4. Beam element with an inside debonded composite patch.

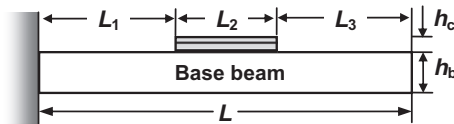


Fig. 5. Cantilevered beam with a surface-bonded composite patch.

Notice that the superscripts R in Eqs. (51) and (52) are used for a finite composite patched beam element with a right-edge debonding.

Similarly, the spectral element equation for a finite composite patched beam element with a right-edge debonding (denoted by the superscript L) can be also derived in the form as

$$\mathbf{S}^L(\omega)\mathbf{d}^L = \mathbf{f}^L \quad (54)$$

The spectral element model for the finite composite patched beam elements with an inside debonding can be then readily derived by simply assembling the spectral element model for right-edge debonding (Eq. (51)) and the spectral element model for left-edge debonding (Eq. (54)) as shown skeptically in Fig. 4.

5. Numerical examples and discussion

5.1. Evaluation of spectral element model

A thorough literature survey shows that exact theoretical or numerical solutions are not available in the literature for the types of composite patched beams considered in this study. Thus, the accuracy of the present spectral element model was evaluated by comparison with solutions obtained using the conventional finite element model developed separately in this study (see Appendix C) and the commercial finite element (FE) analysis package ANSYS [28]. To obtain the ANSYS results presented in this study, a 2-D FE analysis was conducted using eight-node quadratic plane stress elements.

Fig. 5 shows a cantilevered metallic beam with a surface-bonded composite patch. This structure was used as the first example problem for the evaluation of the present spectral element model. The geometric properties of the composite patched beam are $L_1 = L_2 = L_3 = 100$ mm, $b = 20$ mm, $h_b = 2$ mm, and $h_c = 0.5$ mm. The material properties of the metallic base beam are $E = 69$ GPa, $\nu = 0.33$, and $\rho_b = 2700$ kg/m³. The composite patch has an asymmetric cross-ply lay-up $[0/90]_2$, and each ply is made of AS/3501-6 graphite-epoxy material whose material properties are $E_1 = 144.48$ GPa, $E_2 = 9.632$ GPa, $E_3 = 9.412$ GPa, $G_{12} = 4.128$ GPa, $G_{13} = 7.457$ GPa, $G_{23} = 6.516$ GPa, $\nu_{12} = \nu_{13} = \nu_{23} = 0.3$, and $\rho_c = 1389.2$ kg/m³.

Table 1 compares the natural frequencies obtained by the present SEM with those from the conventional FEM and the ANSYS results. Only three finite elements are needed for the SEM results. However, for the FEM analysis, the total number of finite elements was increased until sufficiently converged results are obtained. It is obvious from Table 1 that the FEM results converged to the SEM results as the total number of finite elements was increased; for instance, up to more than 1,000 elements were needed to meet five digit accuracy for the tenth natural frequency. ANSYS results were obtained by using a total of 41,600 finite elements: 38,400 elements for the base beam layer (size: 0.125 mm $\times$ 0.125 mm) and 3,200 elements for the composite patch layer (size: 0.125 mm $\times$ 0.125 mm). The ANSYS results are found to be very

Table 1
Natural frequencies (Hz) of the cantilevered composite patched beam shown in Fig. 5.

Mode	SEM (n)	FEM (n)					ANSYS
	n = 3	n = 3	n = 30	n = 300	n = 600	n = 1500	n = 41,600
1	18.766	18.855	18.777	18.768	18.767	18.766	18.727
2	132.21	133.01	132.43	132.25	132.23	132.21	130.80
3	336.81	367.49	337.45	336.93	336.86	336.81	335.60
4	686.99	807.60	687.22	686.99	686.99	686.99	681.55
5	1145.6	2142.8	1148.3	1146.0	1145.7	1145.6	1137.2
10	4265.2	114,870	4271.7	4265.6	4265.4	4265.2	4265.0
20	16,607	–	16,952	16,613	16,609	16,608	16,511

Note: (n) = the total number of finite elements used in the analysis.

close to the SEM results, and small deviations are probably due to differences in the constitutive relations and kinematic assumptions adopted in the SEM and the ANSYS formulations.

Fig. 6 compares the frequency response functions (FRFs) obtained by the present SEM with those by FEM and ANSYS for the cantilevered composite patched beam. A transverse unit impulse was applied at the free end of the metallic base beam, and the FRFs were obtained from the transverse displacements at the free end excitation point. Fig. 6 shows that the FRF obtained by the three-element SEM is very close to the ANSYS result, where 41,600 finite elements were used. Fig. 6 also shows that the FEM results converged to the SEM results as the total number of finite elements used in the finite element analysis was increased. The ANSYS results are found to be accurate enough so that they are very close to the SEM results at the lower frequency range (see Fig. 6(a)). However, they are found to be less accurate at the higher frequency range (see Fig. 6(b)), and more finite elements are required to improve the accuracy.

For the evaluation of the present spectral element model, a cantilevered composite patched beam, as shown in Fig. 7, was used for the second example problem. A PZT actuator and a PZT sensor were placed on the left-side and right-side of the composite patch, respectively. The composite patched beam has the following dimensions: $L_1 = L_5 = 900$ mm, $L_2 = L_3 = L_4 = 300$ mm, $L_p = 5$ mm, $b = 20$ mm, $h_b = 2$ mm, and $h_p = h_c = 0.5$ mm. The composite patch has the asymmetric cross-ply lay-up $[0/90]_2$, and each ply is made up of AS/3501-6 graphite-epoxy material. The material properties of the metallic base beam and the composite laminate are identical to those used for the first example problem. A Morlet wave signal with a 100 kHz center frequency and having five cycles with peak-to-peak voltage of 200 V was applied to the PZT actuator. The guided waves to be measured by the PZT sensor were predicted by using the SEM, the FEM, and by ANSYS, and the results are compared in Fig. 8. Fig. 8 shows that the guided wave predicted by the FEM converges to that predicted by the present SEM as the total

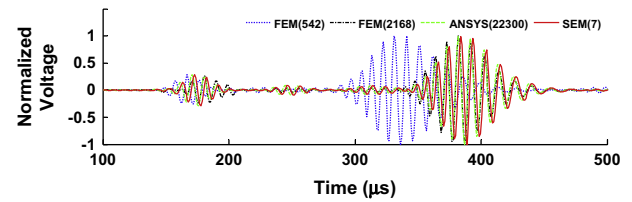


Fig. 8. Simulation of guided waves in the cantilevered composite patched beam shown in Fig. 7.

number of finite elements used in the finite element analysis was increased to more than 2168. The SEM result can be achieved using ANSYS when more than 22,300 finite elements are used. The ANSYS results were obtained with a total of 22,300 finite elements: 21,680 elements for the base beam layer (size: 0.5 mm $\times$ 0.5 mm), 600 elements for the composite patch layer (size: 0.5 mm $\times$ 0.5 mm), and 20 elements for the PZT layer (size: 0.5 mm $\times$ 0.5 mm). In Fig. 8, the first signal represents the symmetric longitudinal wave (S_0 mode), the last largest signal represents the asymmetric transverse wave (A_0 mode), and the second and third signals (between the first and last signals) represent symmetric and asymmetric waves generated by wave mode conversions, respectively. Fig. 8 also shows that in the FEM analysis, a much larger number of finite elements must be employed to predict the accurate location of the A_0 mode than to predict that of the S_0 mode. This and the aforementioned investigations validate the extremely high accuracy of the present spectral element model for composite patched beams. Finally, Table 2 compares the CPU times required to simulate the guided waves displayed in Fig. 8 by using the SEM, the FEM, and ANSYS on a laboratory desktop computer (Intel® Core™ i7-2600 CPU at 3.4 GHz). Table 2 shows that the present SEM requires very low computation time compared the times required by the FEM and ANSYS.

5.2. Effects of the composite patch on the dispersion curves

Fig. 9(a) shows the frequency-wave number and frequency-group velocity dispersion curves for a bare metallic base beam. Fig. 9(b) shows the frequency-wave number and frequency-group velocity dispersion curves for the same metallic base beam that is fully covered with a surface-bonded composite patch. By comparing Fig. 9(a) with Fig. 9(b), we note that (1) additional wave modes, such as the shear and lateral contraction wave modes, are generated in the composite patch layer due to the existence of the composite patch; (2) the lateral contraction wave in the metallic base beam seems to be strongly influenced by the shear wave in the composite patch layer; (3) the cut-off frequencies seem to shift toward the lower frequencies due to the existence of the composite patch; and (4) in general, the dispersion characteristics seem to change significantly at the high frequency (e.g., at about 1 MHz for the present example problem).

5.3. Effects of the composite patch on the guided waves

To investigate the effects of the surface-bonded composite patch on guided wave propagation, we reconsidered the

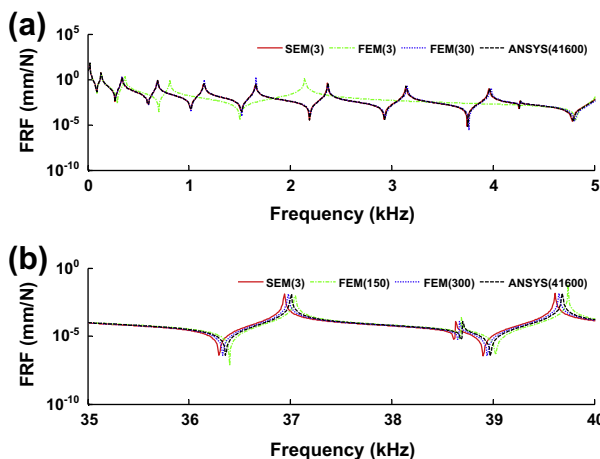


Fig. 6. Frequency response functions for the transverse vibration of the cantilevered composite patched beam shown in Fig. 5.

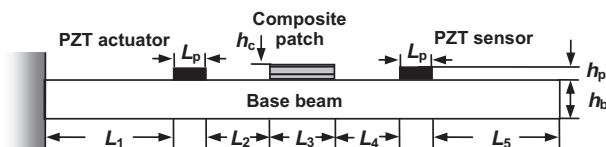


Fig. 7. Cantilevered beam with a surface-bonded composite patch considered for the simulation of guided waves.

Table 2
CPU times required to simulate the waves shown in Fig. 8.

Analysis method	SEM (n)	FEM (n)	ANSYS
	n = 5	n = 542	n = 2168
		n = 22,300	
CPU time (s)	38	1325	15,353
			12,174

Note: (n) = the total number of finite elements used in the analysis.

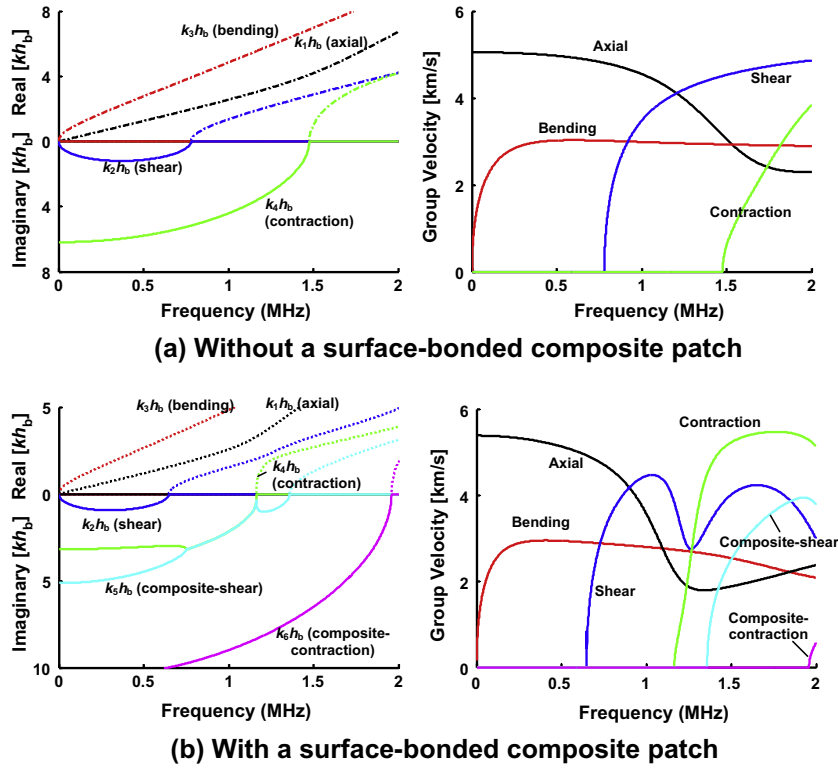


Fig. 9. Comparison of the dispersion curves of the beams without and with a bonded composite patch.

cantilevered metallic base beam shown in Fig. 7. Guided wave simulations were conducted for two cases: (1) without the composite patch, and (2) with the composite patch. Fig. 10 displays the guided wave simulation results. When no composite patch is bonded on the surface of the metallic base beam, only two waves exist: one is the symmetric longitudinal wave (denoted by S_0) and the other is the asymmetric transverse wave (denoted by A_0). As the group velocity of the S_0 wave is faster than that of A_0 wave, Fig. 10 shows that the S_0 wave signal appears earlier than the A_0 wave. However, when a composite patch is placed on the surface of the metallic base beam, Fig. 10 shows that two additional waves occur between the original S_0 wave and the A_0 wave. They are generated by the wave mode conversion when the original S_0 wave and A_0 wave propagate through the composite patch. The composite patch tends to increase the group velocities slightly at the 100 kHz excitation frequency, as shown in Fig. 9. Thus, the waves in the composite patched beam appear slightly earlier, as shown in Fig. 10.

To investigate the effects of the lay-up of a composite patch on guided wave propagation, eight different lay-ups were considered: the unidirectional lay-up $[0]_4$, the symmetric cross-ply lay-up $[0/90]_s$, the asymmetric cross-ply lay-up $[0/90]_2$, the symmetric angle-ply lay-up $[30/-30]_s$, the asymmetric angle-ply lay-up $[30/-30]_2$, the angle-ply symmetric lay-up $[45/-45]_s$, the asymmetric angle-ply lay-up $[45/-45]_2$, and the quasi-isotropic lay-up $[90/$

$45/0/-45]$. Fig. 11(a) compares the guided waves for the symmetric and asymmetric cross-ply lay-ups with those for the unidirectional lay-up. Similarly, Fig. 11(b)–(d) compares guided waves for

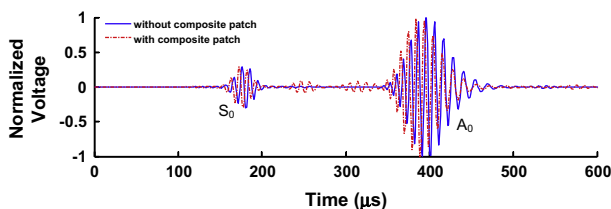


Fig. 10. Effects of the existence of a surface-bonded composite patch on the guided waves.

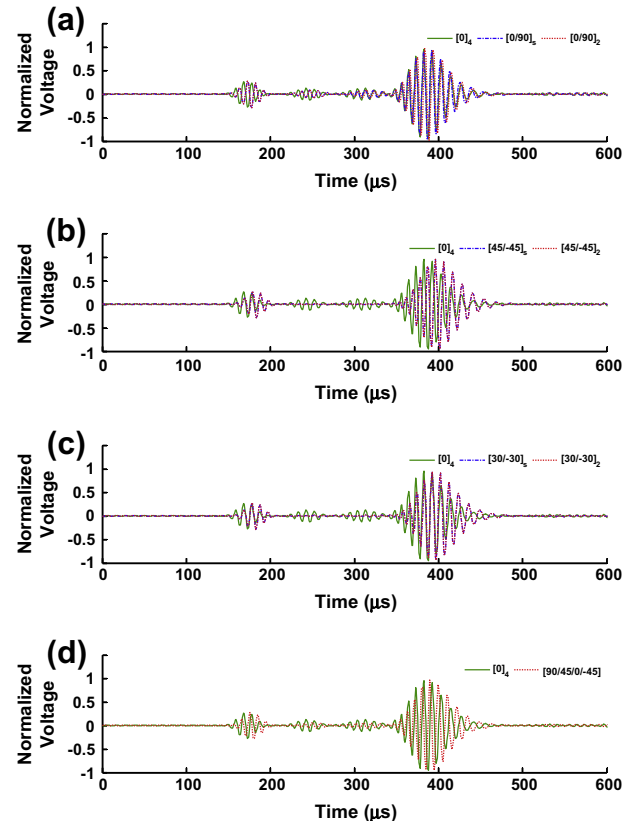


Fig. 11. Effects of the composite patch lay-up on the guided waves.

the cross-ply lay-ups and the quasi-isotropic lay-up, respectively, with those of the unidirectional lay-up. From Fig. 11, we note that angle-ply lay-ups induce the most significant phase lags of the guided waves and generate the weakest waves via mode conversion, when compared with other types of lay-ups.

Fig. 12 shows a metallic beam with a surface bonded composite patch. We considered this beam in order to investigate the guided wave propagations and reflections at the left (free edge) and right (clamped edge) boundaries, and at the PZT actuator and the composite patch. The composite patched beam has the following dimensions: $L_1 = 900$ mm, $L_2 = 300$ mm, $L_3 = 300$ mm, $L_4 = 1205$ mm, $L_p = 5$ mm, $b = 20$ mm, $h_b = 2$ mm, and $h_p = h_c = 0.5$ mm. A Morlet wave signal with a 100 kHz center frequency having five cycles with a peak-to-peak voltage of 200 V was applied to the PZT actuator. Axial (longitudinal) and flexural bending waves propagating in the metallic base beam are shown in Fig. 13(a) and (b), respectively. Fig. 13(c) and (d) display the time histories at $x = 1805$ mm of the composite patched beam. We note that the waves excited by the PZT actuator started from $x = L_1$ (the location of the PZT actuator) and propagated toward the free edge and clamped edge. During the propagation, the waves were reflected at the structural boundaries, the PZT layer, and the composite patch. The reflection locations can be readily identified by using the group velocities of each propagating wave. Comparing Fig. 13(a) and (b), we note that the flexural bending wave is more dispersive than the axial wave because its wave shape spreads out more broadly during propagation.

5.4. Effects of composite patch debonding on natural frequencies

Fig. 14 shows a cantilevered metallic beam with a surface-bonded composite patch whose right edge is partially debonded. The geometric properties are $L_1 = L_2 = L_3 = 100$ mm, $b = 20$ mm, $h_b = 2$ mm, and $h_c = 0.5$ mm. The composite patch has the cross-ply asymmetric lay-up $[0/90]_2$ and each ply is made up of

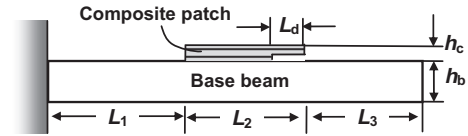


Fig. 14. Cantilevered metallic beam with a right-edge debonded composite patch.

AS/3501-6 graphite-epoxy material. The material properties for the metallic base beam and composite patch are identical to those used for the previous example problems. Table 3 shows the natural frequencies of the example composite patched beam, depending on the size of the debond as represented by the ratio $r = L_d/L_2$. Thus, $r = 0$ means that the composite patch is perfectly bonded on the base beam, while $r = 1$ means that the composite patch is completely debonded from the base beam. As the size of the debond increases, the effective surface-bonded length of the composite patch decreases. This results in weakening the strength of the composite patched beam. Accordingly, Table 3 shows that the natural frequencies of the composite patched beam decrease as the size of the debond increases.

5.5. Effects of composite patch debonding on guided waves

To investigate the effects of composite patch debonding on guided wave propagation, we revisit the cantilevered composite patched beam shown in Fig. 7. Assume that the right edge of the surface-bonded composite patch is debonded, as shown in Fig. 14, and the degree of debonding is represented by the ratio

Table 3

Natural frequencies (Hz) of the metallic beam with a right-edge debonded composite patch.

Mode	$r = L_d/L_2$				
	0	0.25	0.5	0.75	1
1	18.766	18.704	18.580	18.365	18.147
2	132.21	126.08	118.63	118.52	113.70
3	336.81	326.23	325.39	323.51	318.25
4	686.99	677.79	668.32	657.91	623.32
5	1145.6	1121.7	1100.5	1072.8	1029.7
10	4265.2	4062.1	4007.5	3915.7	3886.5
20	16,607	16,344	15,931	15,723	15,238

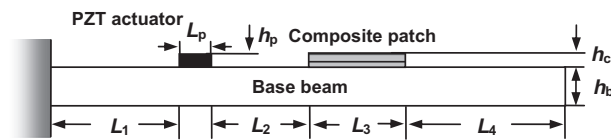


Fig. 12. Cantilevered metallic beam with a composite patch and PZT actuator.

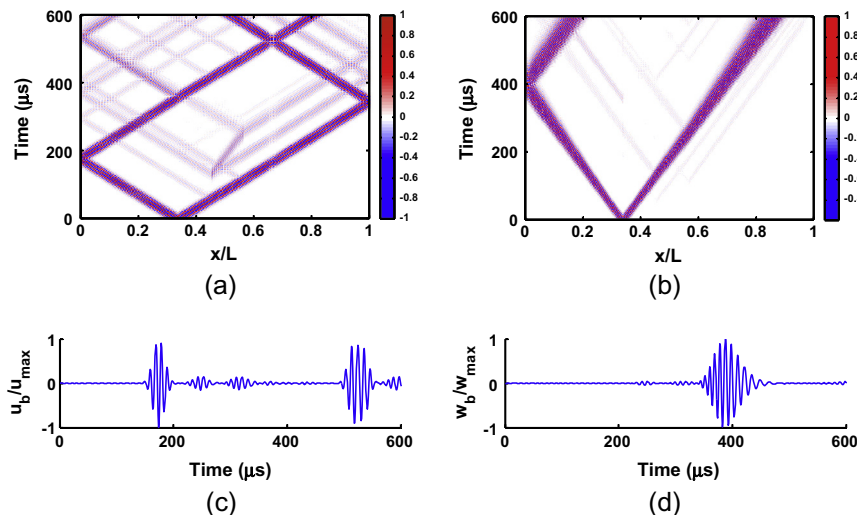


Fig. 13. Axial and flexural dynamic responses (waves) in the smart metallic beam: (a) $u_b(x, 0, t)$, (b) $w_b(x, 0, t)$, (c) $u_b(1805, 0, t)$, (d) $w_b(1805, 0, t)$.

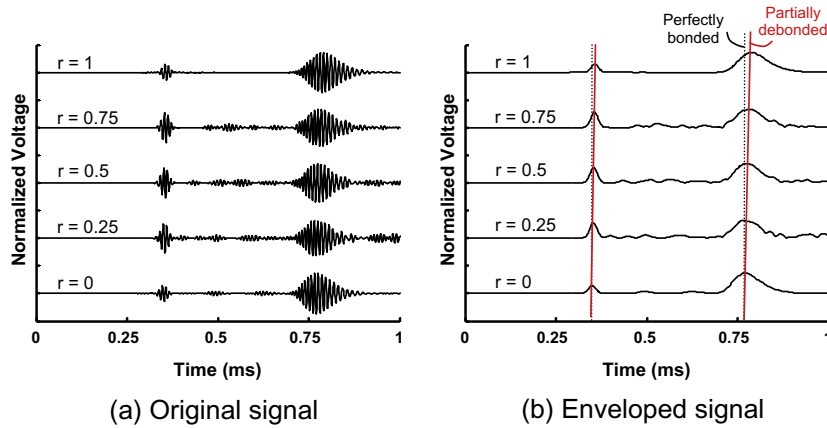


Fig. 15. Effects of debonding length ratio r on guided waves.

$r = L_d/L_2$. The composite patch has an asymmetric cross-ply lay-up $[0/90]_2$. The material properties for the metallic base beam and composite patch are identical to those used for the previous example problems. Fig. 15 shows the guided waves for various values of r . As discussed in Section 5.3, the surface-bonded composite patch has been found to increase the group velocities of the guided waves, especially those excited at 100 kHz. Thus, the partial debonding of the composite patch slightly decreases the group velocities of the guided waves in a reverse way. Accordingly, Fig. 15 shows that as the size of the right-edge debond increases (i.e., as r is increased), the wave peaks tend to be delayed more. The degree of the time delay could be a useful feature in identifying (detecting and locating) the debonds generated in a surface-bonded composite patch.

6. Conclusions

This paper presents a frequency domain spectral element model for the guided waves in the metallic beams with a bonded composite patch. The spectral element model is formulated from six coupled equations of motion which are derived by adopting Timoshenko beam theory and Mindlin–Herrmann rod theory for the transverse bending motion and the longitudinal motion, respectively. Through the numerical analysis, the effects of the composite patch and its debonding were investigated as follows:

- (1) Additional dispersion curves for the shear and lateral contraction waves in the composite patch layer appear.
- (2) The lateral contraction wave in the base beam is more strongly influenced by the shear wave in the composite patch layer.
- (3) Dispersion curves change more significantly at higher frequencies and the cut-off frequencies shift to the lower frequencies.
- (4) Composite patch generates additional wave modes via the mode conversions at the end edges of a composite patch.
- (5) Angle-ply lay-up seems to induce the most significant phase lags of the guided waves, and generate the weakest mode conversion-induced waves when compared with other types of lay-ups.
- (6) Natural frequencies of a composite patched beam decrease as the size of the debond increases.
- (7) Time-of-arrivals of the guided waves are delayed as the size of the debond increases, which can be used as a useful information for detecting debonds generated in a composite patched beam.

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Appendix A. Base beam

A.1. Stress–strain relations

By adopting Mindlin–Herrmann rod theory [25] for beams made of isotropic materials, the stress–strain relations for the axial vibrations of isotropic beams can be reduced from the stress–strain relations of three-dimensional (3-D) isotropic materials by imposing the conditions $\sigma_{yy} = \tau_{xy} = \tau_{yz} = 0$ as follows:

$$\begin{Bmatrix} \sigma_{xx}^{(a)} \\ \sigma_{zz}^{(a)} \\ \tau_{xz}^{(a)} \end{Bmatrix} = \begin{bmatrix} C_{11}^* & C_{13}^* & 0 \\ C_{13}^* & C_{11}^* & 0 \\ 0 & 0 & C_{55}^* \end{bmatrix} \begin{Bmatrix} \varepsilon_{xx}^{(a)} \\ \varepsilon_{zz}^{(a)} \\ \gamma_{xz}^{(a)} \end{Bmatrix} \quad (A1)$$

where

$$C_{11}^* = \frac{E}{1-\nu^2}, \quad C_{13}^* = \frac{E\nu}{1-\nu^2}, \quad C_{55}^* = G \quad (A2)$$

and where E is Young's modulus, ν is Poisson's ratio, and G is the shear modulus.

Similarly, by adopting Timoshenko beam theory [24] for transverse bending vibration, the stress–strain relations for the transverse bending vibrations of isotropic beams can be reduced from those of three-dimensional (3-D) isotropic materials by imposing the conditions $\sigma_{yy} = \tau_{xy} = \tau_{yz} = 0$ and $\sigma_{zz} = 0$ as follows:

$$\begin{Bmatrix} \sigma_{xx}^{(b)} \\ \tau_{xz}^{(b)} \end{Bmatrix} = \begin{bmatrix} Q_{11}^* & 0 \\ 0 & Q_{55}^* \end{bmatrix} \begin{Bmatrix} \sigma_{xx}^{(b)} \\ \gamma_{xz}^{(b)} \end{Bmatrix} \quad (A3)$$

where

$$Q_{11}^* = E, \quad Q_{55}^* = G \quad (A4)$$

A.2. Strains

The strains in the base beam can be obtained from the displacement fields given by Eq. (1):

$$\begin{aligned} \varepsilon_{xx} &= \partial u_b / \partial x = \varepsilon_{xx}^{(a)} + \varepsilon_{xx}^{(b)}, \quad \varepsilon_{zz} = \partial w_b / \partial z = \varepsilon_{zz}^{(a)} \\ \gamma_{xz} &= \partial u_b / \partial z + \partial w_b / \partial x = \gamma_{xz}^{(a)} + \gamma_{xz}^{(b)} \end{aligned} \quad (A5)$$

where

$$\begin{aligned} \varepsilon_{xx}^{(a)} &= u'_{b0}, \quad \varepsilon_{zz}^{(a)} = \psi_b, \quad \gamma_{xz}^{(a)} = z\psi'_b, \quad \varepsilon_{xx}^{(b)} = -z\theta'_b, \quad \gamma_{xz}^{(b)} \\ &= (w'_{b0} - \theta_b) \end{aligned} \quad (A6)$$

A.3. Strain energy

By using Eqs. (A1), (A3) and (A6), the strain energy in the base beam of length L can be obtained as follows:

$$\begin{aligned} V_b &= V_b^{(a)} + V_b^{(b)} \\ &= \frac{1}{2} \int_0^L (\sigma_{xx}^{(a)} \varepsilon_{xx}^{(a)} + \sigma_{zz}^{(a)} \varepsilon_{zz}^{(a)} + \tau_{xz}^{(a)} \gamma_{xz}^{(a)}) dA_b dx + \frac{1}{2} \\ &\quad \times \int_0^L (\sigma_{xx}^{(b)} \varepsilon_{xx}^{(b)} + \tau_{xz}^{(b)} \gamma_{xz}^{(b)}) dA_b dx \end{aligned} \quad (A7)$$

A.4. Resultant forces and moments

The resultant forces and moments in the base beam are defined as

$$(N_b, Q_b, M_b, R_b) = b \int_{-h_b/2}^{h_b/2} (\sigma_{xx}^{(a)}, \kappa_b \tau_{xz}^{(b)}, -\sigma_{xx}^{(b)} z, \kappa_b \tau_{xz}^{(a)} z) dz \quad (A8)$$

where κ_b is the shear correction factor of the base beam. By using Eqs. (A1), (A3), and (A6), the resultant forces and moments can be obtained from Eq. (A8):

$$\begin{aligned} N_b &= A_b C_{11}^* u'_{b0} + A_b C_{13}^* \psi_b, \quad Q_b = \kappa_b Q_{55}^* A_b (w'_{b0} - \theta_b) \\ M_b &= I_{ob} Q_{11}^* \theta'_b, \quad R_b = \kappa_b C_{55}^* I_{ob} \psi'_b \end{aligned} \quad (A9)$$

where

$$A_b = b \int_{-h_b/2}^{h_b/2} z dz = b h_b, \quad I_{ob} = b \int_{-h_b/2}^{h_b/2} z^2 dz = \frac{1}{12} b h_b^3 \quad (A10)$$

Appendix B. Composite patch

B.1. Stress–strain relations

Similar to the isotropic base beam in Appendix A, the stress–strain relations for the axial vibration of a one-dimensional (1-D) composite patch can be reduced from those for the 3-D orthotropic materials as follows:

$$\begin{Bmatrix} \sigma_{xx}^{(a)} \\ \sigma_{zz}^{(a)} \\ \tau_{xz}^{(a)} \end{Bmatrix} = \begin{bmatrix} \bar{C}_{11}^* & \bar{C}_{13}^* & 0 \\ \bar{C}_{13}^* & \bar{C}_{33}^* & 0 \\ 0 & 0 & \bar{C}_{55}^* \end{bmatrix} \begin{Bmatrix} \varepsilon_{xx}^{(a)} \\ \varepsilon_{zz}^{(a)} \\ \gamma_{xz}^{(a)} \end{Bmatrix} \quad (B1)$$

For the transverse bending vibration, we obtain

$$\begin{Bmatrix} \sigma_{xx}^{(b)} \\ \tau_{xz}^{(b)} \end{Bmatrix} = \begin{bmatrix} \bar{Q}_{11}^* & 0 \\ 0 & \bar{Q}_{55}^* \end{bmatrix} \begin{Bmatrix} \sigma_{xx}^{(b)} \\ \gamma_{xz}^{(b)} \end{Bmatrix} \quad (B2)$$

In Eq. (B1), $\bar{C}_{ij}^*$ are the transformed elastic stiffness coefficients defined in Ref. [23] and $\bar{Q}_{ij}^*$ in Eq. (B2) are defined as

$$\bar{Q}_{11}^* = \bar{C}_{11}^* - \frac{\bar{C}_{13}^{*2}}{\bar{C}_{33}^*}, \quad \bar{Q}_{55}^* = \bar{C}_{55}^* \quad (B3)$$

B.2. Strains

The strains in the composite patch can be obtained from the displacement fields given by Eq. (2) as

$$\begin{aligned} \varepsilon_{xx} &= \partial u_c / \partial x = \varepsilon_{xx}^{(a)} + \varepsilon_{xx}^{(b)}, \quad \varepsilon_{zz} = \partial w_c / \partial z = \varepsilon_{zz}^{(a)} \\ \gamma_{xz} &= \partial u_c / \partial z + \partial w_c / \partial x = \gamma_{xz}^{(a)} + \gamma_{xz}^{(b)} \end{aligned} \quad (B4)$$

where

$$\varepsilon_{xx}^{(a)} = u'_{c0}, \quad \varepsilon_{zz}^{(a)} = \psi_c, \quad \gamma_{xz}^{(a)} = z\psi'_c, \quad \varepsilon_{xx}^{(b)} = -z\theta'_c, \quad \gamma_{xz}^{(b)} = (w'_{c0} - \theta_c) \quad (B5)$$

B.3. Strain energy

By using Eqs. (B1), (B2) and (B5), the strain energy in the composite patch of length L is given by

$$\begin{aligned} V_c &= V_c^{(a)} + V_c^{(b)} = \frac{1}{2} \int_0^L (\sigma_{xx}^{(a)} \varepsilon_{xx}^{(a)} + \sigma_{zz}^{(a)} \varepsilon_{zz}^{(a)} + \tau_{xz}^{(a)} \gamma_{xz}^{(a)}) dA_c dx \\ &\quad + \frac{1}{2} \int_0^L (\sigma_{xx}^{(b)} \varepsilon_{xx}^{(b)} + \tau_{xz}^{(b)} \gamma_{xz}^{(b)}) dA_c dx \end{aligned} \quad (B6)$$

B.4. Resultant forces and moments

The resultant forces and moments in the composite patch are defined as

$$(N_c, Q_c, M_c, R_c) = b \int_{-h_c/2}^{h_c/2} (\sigma_{xx}^{(a)}, \kappa_c \tau_{xz}^{(b)}, -\sigma_{xx}^{(b)} z, \kappa_c \tau_{xz}^{(a)} z) dz \quad (B7)$$

where κ_c is the shear correction factor of the composite patch. By using Eqs. (B1), (B2), and (B4), the resultant forces and moments can be obtained from Eq. (B7) as follows:

$$\begin{aligned} N_c(x, t) &= b A_{11} u'_{c0} + b A_{13} \psi_c, \quad Q_c(x, t) = \kappa_c b \bar{A}_{55} (w'_{c0} - \theta_b) \\ M_c(x, t) &= b \bar{D}_{11} \theta'_c, \quad R_c(x, t) = \kappa_c b D_{55} \psi'_c \end{aligned} \quad (B8)$$

where

$$\begin{aligned} (A_{ij}, D_{ij}) &= \sum_{k=1}^N \int_{z_{k-1}}^{z_k} \bar{C}_{ij}^{*(k)} (1, z^2) dz \\ (\bar{A}_{ij}, \bar{D}_{ij}) &= \sum_{k=1}^N \int_{z_{k-1}}^{z_k} \bar{Q}_{ij}^{*(k)} (1, z^2) dz \end{aligned} \quad (B9)$$

Appendix C. Finite element model for the composite patched beam

A finite element model for the composite patched Timoshenko beam can be derived by assuming the following displacement fields:

$$\begin{aligned} u_{b0}(x, t) &= \mathbf{N}_u(x) \mathbf{d}(t), \quad w_{b0}(x, t) = \mathbf{N}_w(x) \mathbf{d}(t) \\ \theta_b(x, t) &= \mathbf{N}_{\theta_b}(x) \mathbf{d}(t), \quad \psi_b(x, t) = \mathbf{N}_{\psi_b}(x) \mathbf{d}(t) \\ \theta_c(x, t) &= \mathbf{N}_{\theta_c}(x) \mathbf{d}(t), \quad \psi_c(x, t) = \mathbf{N}_{\psi_c}(x) \mathbf{d}(t) \end{aligned} \quad (C1)$$

where $\mathbf{d}(t)$ is the nodal DOFs vector and $\mathbf{N}_u(x)$, $\mathbf{N}_w(x)$, $\mathbf{N}_{\theta_b}(x)$, $\mathbf{N}_{\psi_b}(x)$, $\mathbf{N}_{\theta_c}(x)$, and $\mathbf{N}_{\psi_c}(x)$ are the shape function matrices, which are defined as

$$\mathbf{d}(t) = \{u_{b01} \ w_{b01} \ \theta_{b1} \ \psi_{b1} \ \theta_{c1} \ \psi_{c1} \ u_{b02} \ w_{b02} \ \theta_{b2} \ \psi_{b2} \ \theta_{c2} \ \psi_{c2}\}^T \quad (C2)$$

$$\begin{aligned} \mathbf{N}_u &= [N_{u1} \ 0 \ 0 \ 0 \ 0 \ 0 \ N_{u2} \ 0 \ 0 \ 0 \ 0 \ 0] \\ \mathbf{N}_w &= [0 \ N_{w1} \ N_{w2} \ 0 \ 0 \ 0 \ 0 \ N_{w3} \ N_{w4} \ 0 \ 0 \ 0] \\ \mathbf{N}_{\theta_b} &= [0 \ N_{\theta_b1} \ N_{\theta_b2} \ 0 \ 0 \ 0 \ 0 \ N_{\theta_b3} \ N_{\theta_b4} \ 0 \ 0 \ 0] \\ \mathbf{N}_{\psi_b} &= [0 \ 0 \ 0 \ N_{\psi_b1} \ 0 \ 0 \ 0 \ 0 \ 0 \ N_{\psi_b1} \ 0 \ 0] \\ \mathbf{N}_{\theta_c} &= [0 \ 0 \ 0 \ 0 \ N_{\theta_c1} \ 0 \ 0 \ 0 \ 0 \ 0 \ N_{\theta_c2} \ 0] \\ \mathbf{N}_{\psi_c} &= [0 \ 0 \ 0 \ 0 \ 0 \ N_{\psi_c1} \ 0 \ 0 \ 0 \ 0 \ 0 \ N_{\psi_c2}] \end{aligned} \quad (C3)$$

where

$$\begin{aligned} N_{u1} &= l^{-1}(l-x), & N_{u2} &= l^{-1}x \\ N_{w1} &= \alpha l^{-3}(l-x)(lx-2x^2+l^2+12\beta l^2), & N_{w2} &= \alpha l^{-2}x(l-x)(l-x+6l\beta) \\ N_{w3} &= \alpha l^{-3}x(3lx-2x^2+12l^2\beta), & N_{w4} &= -\alpha l^{-2}x(l-x)(x+6l\beta) \\ N_{\theta b1} &= 6\alpha x(l-x), & N_{\theta b2} &= \alpha(l-x)(l^2+12\beta-3lx) \\ N_{\theta b3} &= 6\alpha x(l-x), & N_{\theta b4} &= \alpha x(-2l^2+12\beta+3lx) \\ N_{\psi b1} &= l^{-1}(l-x), & N_{\psi b2} &= l^{-1}x \\ N_{\theta c1} &= l^{-1}(l-x), & N_{\theta c2} &= l^{-1}x \\ N_{\psi c1} &= l^{-1}(l-x), & N_{\psi c2} &= l^{-1}x \end{aligned} \quad (C4)$$

and where

$$\alpha = \frac{1}{1+12\beta}, \quad \beta = \frac{I_{ob}Q_{11}^*}{\kappa Q_{55}^*A_b} \quad (C5)$$

Substituting Eq. (C1) into the weak form of the governing equations of motion, Eq. (2.16), gives the finite element equation in the following form:

$$\mathbf{M}\ddot{\mathbf{d}} + \mathbf{K}\mathbf{d} = \mathbf{f}_d + \mathbf{f}_c \quad (C6)$$

where

$$\begin{aligned} \mathbf{M} &= \int_0^l \left\{ m_{11}\mathbf{N}_u^T\mathbf{N}_u + m_{22}\mathbf{N}_w^T\mathbf{N}_w + m_{33}\mathbf{N}_{\theta b}^T\mathbf{N}_{\theta b} + m_{44}\mathbf{N}_{\psi b}^T\mathbf{N}_{\psi b} + m_{55}\mathbf{N}_{\theta c}^T\mathbf{N}_{\theta c} \right. \\ &\quad - m_{66}\mathbf{N}_{\psi c}^T\mathbf{N}_{\psi c} - m_{13}\left(\mathbf{N}_u^T\mathbf{N}_{\theta b} + \mathbf{N}_{\theta b}^T\mathbf{N}_u\right)U_b - m_{15}\left(\mathbf{N}_u^T\mathbf{N}_{\theta c} + \mathbf{N}_{\theta c}^T\mathbf{N}_u\right) \\ &\quad + m_{24}\left(\mathbf{N}_w^T\mathbf{N}_{\psi b} + \mathbf{N}_{\psi b}^T\mathbf{N}_w\right) + m_{26}\left(\mathbf{N}_w^T\mathbf{N}_{\psi c} + \mathbf{N}_{\psi c}^T\mathbf{N}_w\right) \\ &\quad \left. + m_{35}\left(\mathbf{N}_{\theta b}^T\mathbf{N}_{\theta c} + \mathbf{N}_{\theta c}^T\mathbf{N}_{\theta b}\right) + m_{46}\left(\mathbf{N}_{\psi b}^T\mathbf{N}_{\psi c} + \mathbf{N}_{\psi c}^T\mathbf{N}_{\psi b}\right) \right\} dx \end{aligned} \quad (C7)$$

$$\begin{aligned} \mathbf{K} &= \int_0^l \left\{ EA_u\mathbf{N}_u^T\mathbf{N}_u' + \kappa GAN_w^T\mathbf{N}_w' + EI_{\theta b}\mathbf{N}_{\theta b}^T\mathbf{N}_{\theta b}' + \kappa GI_{\psi b}\mathbf{N}_{\psi b}^T\mathbf{N}_{\psi b}' \right. \\ &\quad + EI_{\theta c}\mathbf{N}_{\theta c}^T\mathbf{N}_{\theta c}' + \kappa GI_{\psi c}\mathbf{N}_{\psi c}^T\mathbf{N}_{\psi c}' + \kappa GA_b\mathbf{N}_{\theta b}^T\mathbf{N}_{\theta b} + EA_{\psi b}\mathbf{N}_{\psi b}^T\mathbf{N}_{\psi b} \\ &\quad + \kappa GA_c\mathbf{N}_{\theta c}^T\mathbf{N}_{\theta c} + EA_{\psi c}\mathbf{N}_{\psi c}^T\mathbf{N}_{\psi c} - K_1\left(\mathbf{N}_u^T\mathbf{N}_{\theta b}' + K_1\mathbf{N}_{\theta b}^T\mathbf{N}_u'\right) \\ &\quad - K_2\left(\mathbf{N}_u^T\mathbf{N}_{\theta c}' + \mathbf{N}_{\theta c}^T\mathbf{N}_u'\right) + K_3\left(\mathbf{N}_u^T\mathbf{N}_{\psi b}' + \mathbf{N}_{\psi b}^T\mathbf{N}_u'\right) \\ &\quad + K_4\left(\mathbf{N}_u^T\mathbf{N}_{\psi c}' + \mathbf{N}_{\psi c}^T\mathbf{N}_u'\right) + K_5\left(\mathbf{N}_w^T\mathbf{N}_{\psi b}' + \mathbf{N}_{\psi b}^T\mathbf{N}_w'\right) \\ &\quad + K_6\left(\mathbf{N}_w^T\mathbf{N}_{\psi c}' + \mathbf{N}_{\psi c}^T\mathbf{N}_w'\right) - K_7\left(\mathbf{N}_w^T\mathbf{N}_{\theta b}' + \mathbf{N}_{\theta b}^T\mathbf{N}_w'\right) \\ &\quad - K_8\left(\mathbf{N}_w^T\mathbf{N}_{\theta c}' + \mathbf{N}_{\theta c}^T\mathbf{N}_w'\right) + K_9\left(\mathbf{N}_{\theta b}^T\mathbf{N}_{\theta c}' + \mathbf{N}_{\theta c}^T\mathbf{N}_{\theta b}'\right) \\ &\quad - K_{10}\left(\mathbf{N}_{\theta b}^T\mathbf{N}_{\psi c}' + \mathbf{N}_{\psi c}^T\mathbf{N}_{\theta b}'\right) + K_{11}\left(\mathbf{N}_{\psi b}^T\mathbf{N}_{\psi c}' + \mathbf{N}_{\psi c}^T\mathbf{N}_{\psi b}'\right) \\ &\quad - K_{12}\left(\mathbf{N}_{\psi b}^T\mathbf{N}_{\theta c}' + \mathbf{N}_{\theta c}^T\mathbf{N}_{\psi b}'\right) - K_{13}\left(\mathbf{N}_{\theta c}^T\mathbf{N}_{\psi c}' + \mathbf{N}_{\psi c}^T\mathbf{N}_{\theta c}'\right) \\ &\quad \left. - K_{14}\left(\mathbf{N}_{\theta c}^T\mathbf{N}_{\psi c}' + \mathbf{N}_{\psi c}^T\mathbf{N}_{\theta c}'\right) \right\} dx \end{aligned} \quad (C8)$$

and

$$\begin{aligned} \mathbf{f}_d(t) &= \int_0^l \left\{ f_{bu}(x, t)\mathbf{N}_u^T + f_{bw}(x, t)\mathbf{N}_w^T + f_{b\theta}(x, t)\mathbf{N}_{\theta b}^T \right\} dx \\ \mathbf{f}_c(t) &= \left\{ N_{b1}(t) + N_{c1}(t)Q_{b1}(t) + Q_{c1}(t) - M_{b1}(t) + \frac{1}{2}h_bN_{c1}(t) \right. \\ &\quad \left. \frac{1}{2}h_bQ_{c1}(t) + \frac{1}{2}h_cN_{c1}(t) - M_{c1}(t) \frac{1}{2}h_cQ_{c1}(t) \right. \\ &\quad \left. N_{b2}(t) + N_{c2}(t)Q_{b2}(t) + Q_{c2}(t) + M_{b2}(t) - \frac{1}{2}h_bN_{c2}(t) \right. \\ &\quad \left. \frac{1}{2}h_bQ_{c2}(t) - \frac{1}{2}h_cN_{c2}(t) + M_{c2}(t) \frac{1}{2}h_cQ_{c2}(t) \right\}^T \end{aligned} \quad (C9)$$

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